

Incidence-Guided Treatment Assignment Designs for Spatial Cluster Randomized Trials

Comprehensive Simulation Study Report

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1 Introduction

This report provides a comprehensive reference for the simulation study evaluating **8 treatment assignment designs** for Spatial Cluster Randomized Trials (CRTs) when outcome incidence is heterogeneous and spatially structured.

Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment strategy minimizes estimation error for the direct intervention effect $\hat{\tau}$ across a range of spatial dependence and spillover configurations?

Application context: Sudden Unexpected Death (SUD) in North Carolina counties. The Poisson incidence mode uses an SUD base rate of 35 per 100,000, consistent with county-level analyses.

The study uses Maximum Likelihood Estimation via a spatial autoregressive model (MLE) as the primary estimator. The full simulation has 12,800 scenarios (five values of τ); the primary $\tau = 1.0$ slice analyzed in most of this report has 2,560 scenarios. Queen contiguity is the primary neighbor structure and rook is a sensitivity analysis; **unless a result says otherwise, the tables and figures below pool rook and queen**. Three analytical questions structure the results:

- **Q1:** Which design performs best within each incidence generation mode?
- **Q2:** How do design rankings change as individual parameters vary?
- **Q3:** Which design wins for specific parameter combinations?

2 Study Region and Spatial Structure

The study region is a regular square lattice grid:

$$\mathcal{G} = \{s_1, s_2, \dots, s_N\}$$

where each cluster s_i is located at integer coordinates (x_i, y_i) on a $G \times G$ grid, so $N = G^2$. In this study, $G = 10$ and $N = 100$.

2.1 Spatial Contiguity

Two contiguity definitions are considered:

- **Rook contiguity:** clusters sharing an edge (4-connected).

$$\mathcal{N}_R(s_i) = \{s_j : |x_i - x_j| + |y_i - y_j| = 1\}$$

- **Queen contiguity:** clusters sharing an edge or vertex (8-connected).

$$\mathcal{N}_Q(s_i) = \{s_j : \max(|x_i - x_j|, |y_i - y_j|) = 1\}$$

2.2 Spatial Weight Matrix

Let $\mathbf{A}$ denote the binary adjacency matrix where $A_{ij} = \mathbf{1}[s_j \in \mathcal{N}(s_i)]$. The row-standardized spatial weight matrix is:

$$W_{ij} = \frac{A_{ij}}{\sum_{k=1}^N A_{ik}}, \quad i \neq j; \quad W_{ii} = 0$$

This ensures $\sum_j W_{ij} = 1$ for each row, so spatial lags represent neighborhood averages. We construct $\mathbf{W}_R$ (rook) and $\mathbf{W}_Q$ (queen) weight matrices.

3 Outcome Data Generating Process

3.1 Spatial Durbin Model (SDM)

The outcome vector $\mathbf{Y} = (Y_1, \dots, Y_N)^\top$ is generated from the Spatial Durbin Model:

$$\mathbf{Y} = (\mathbf{I}_N - \rho \mathbf{W})^{-1} [\tau \mathbf{Z} + \gamma \cdot \text{Spill}(\mathbf{Z}) + \beta \mathbf{X} + \varepsilon]$$

where:

Symbol	Description	Role
$\mathbf{Y}$	$N \times 1$ outcome vector	Response
ρ	Spatial autocorrelation parameter	Controls global spatial dependence in $\mathbf{Y}$
$\mathbf{W}$	$N \times N$ row-standardized weight matrix	Spatial structure (rook or queen)
τ	Direct treatment effect	Estimand of interest (swept: $\{0.8, 1.0, 1.5, 2.0, 3.0\}$; primary: 1.0)
$\mathbf{Z}$	$N \times 1$ binary treatment vector	$Z_i \in \{0, 1\}$
γ	Spillover effect magnitude	Indirect effect via treated neighbors
$\text{Spill}(\mathbf{Z})$	$N \times 1$ spillover term	See below
β	Effect of baseline incidence on outcome	Covariate coefficient
$\mathbf{X}$	$N \times 1$ baseline incidence vector	$X_i \in [0, 1]$
ε	$N \times 1$ error vector	$\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

The spatial multiplier $(\mathbf{I}_N - \rho \mathbf{W})^{-1}$ induces global spatial dependence.

Stationarity condition: $|\rho| < 1/\lambda_{\max}(\mathbf{W})$. For row-standardized $\mathbf{W}$, $\lambda_{\max} = 1$, so $|\rho| < 1$.

3.2 Spillover Construction

$$\text{Spill}_i = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|} = (\mathbf{WZ})_i$$

Two spillover types are considered:

1. **Control-only spillover:** $\text{Spill}_i^{(\text{ctrl})} = \gamma \cdot (\mathbf{WZ})_i \cdot (1 - Z_i)$
2. **Bidirectional spillover:** $\text{Spill}_i^{(\text{both})} = \gamma \cdot (\mathbf{WZ})_i$

4 Incidence Generation Modes

The baseline incidence vector $\mathbf{X}$ represents prior observed outcome rates. Three generation modes are considered, all producing values in $[0, 1]$.

4.1 Mode 1: Independent and Identically Distributed (iid)

$$X_i \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1), \quad i = 1, \dots, N$$

No spatial structure; serves as the baseline. One config ($\rho_X = N/A$).

4.2 Mode 2: Spatially Correlated (SAR Filter)

$$\mathbf{X}^* = (\mathbf{I}_N - \rho_X \mathbf{W})^{-1} \mathbf{u}, \quad \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$$

$$X_i = \Phi(X_i^*)$$

where $\Phi(\cdot)$ is the standard normal CDF. Two configs: $\rho_X \in \{0.20, 0.50\}$.

4.3 Mode 3: Poisson Count-Based

1. Spatially correlated log-rates: $\log \lambda^* = \mu_0 \cdot \mathbf{1} + (\mathbf{I}_N - \rho_X \mathbf{W})^{-1} \mathbf{u}$
2. Poisson counts: $C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$ where $P_i = 100,000$ (so $\lambda P \approx 35$ expected deaths per cluster)
3. Observed rates: $R_i = C_i / P_i$
4. Rank-normalized: $X_i = (\text{rank}(R_i) - 0.5) / N$

Base rate $\mu_0 = \log(35/100,000)$ for SUD. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Critical design invariant: Incidence is generated **once per (mode, ρ_X) configuration**: $K = 10$ surfaces $\mathbf{X}_1, \dots, \mathbf{X}_{10}$, held fixed across all $(\rho, \gamma, \text{nb_type})$ scenarios. **Matched surfaces:** each design is drawn from surface $\mathbf{X}_k$, and the outcome and the analysis use the same $\mathbf{X}_k$.

4.4 Incidence Spatial Patterns

5 Treatment Assignment Designs

Eight designs are evaluated. Each produces a binary treatment vector $\mathbf{Z} \in \{0, 1\}^N$.

Incidence Patterns by Generation Mode

Single realization on 10x10 grid

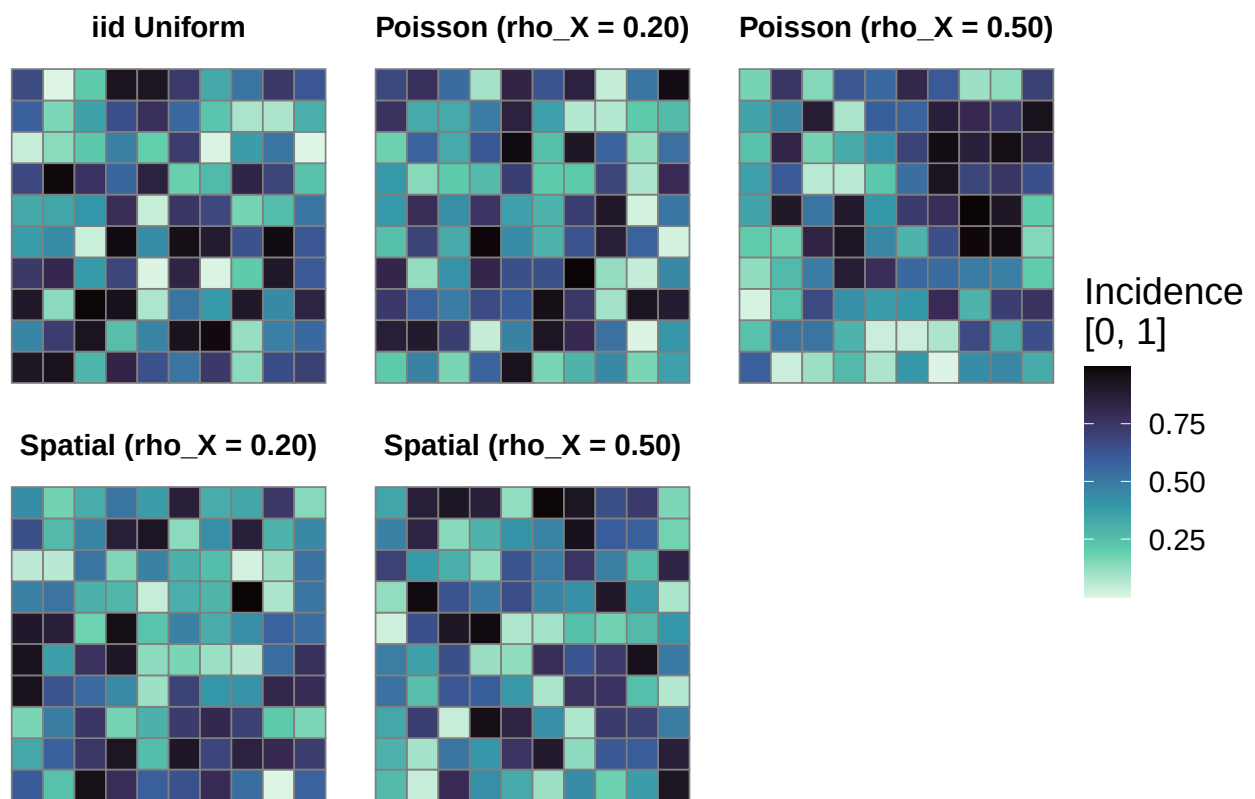


Figure 1: Incidence spatial patterns across generation modes. Color represents incidence (darker = higher burden). Spatial and Poisson modes show visible spatial clustering absent from iid.

Marginal Distribution of Incidence Values by Mode

Pooled across 50 resamples on 10x10 grid

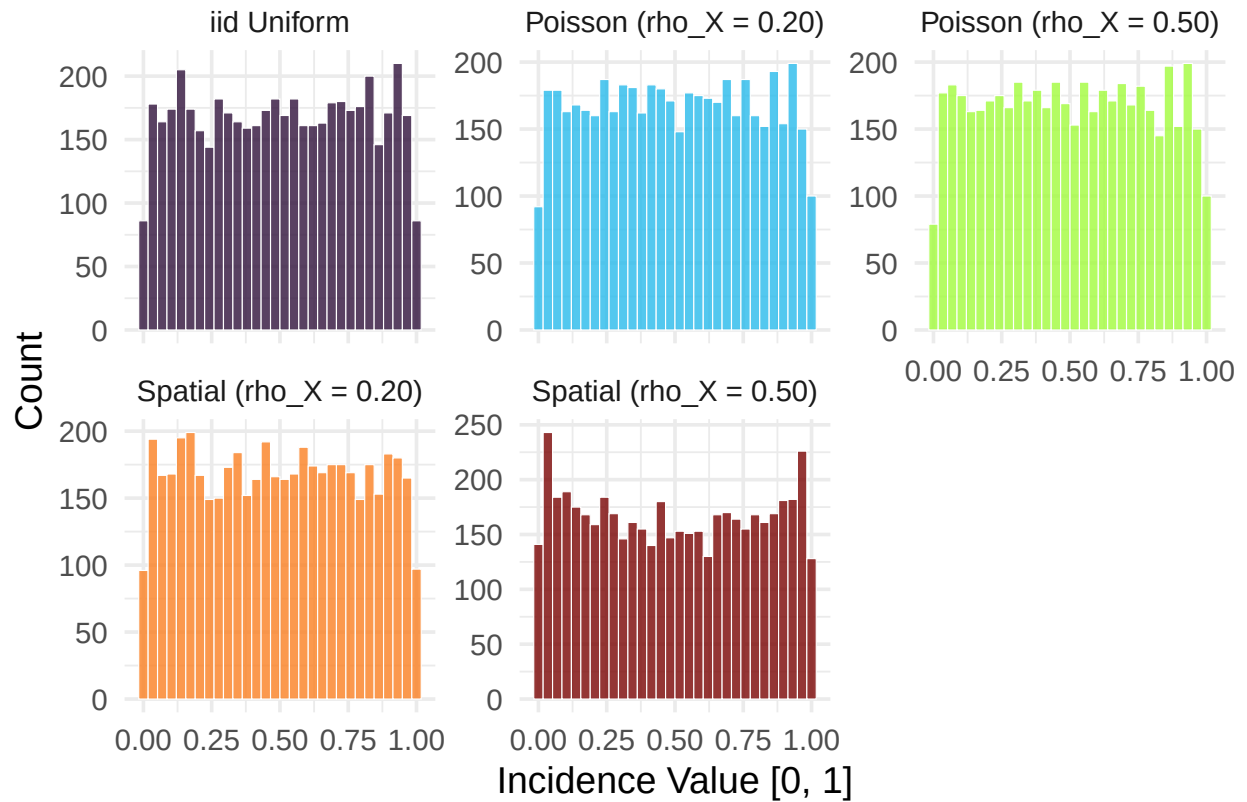


Figure 2: Marginal distribution of incidence values by mode, pooled across 50 resamples. All modes are marginally Uniform on $[0,1]$; spatial structure manifests in spatial covariance, not the marginal.

Table 1: 8 treatment assignment strategies evaluated in this simulation

ID	Name	Deterministic?
1	Design 1: Checkerboard	Yes
2	Design 2: High Incidence Focus	No
3	Design 3: Saturation Quadrants	No
4	Design 4: Isolation Buffer	No
5	Design 5: 2x2 Blocking	No
6	Design 6: Balanced Quartiles	No
7	Design 7: Balanced Halves	No
8	Design 8: Incidence-Guided Saturation Quadrants	No

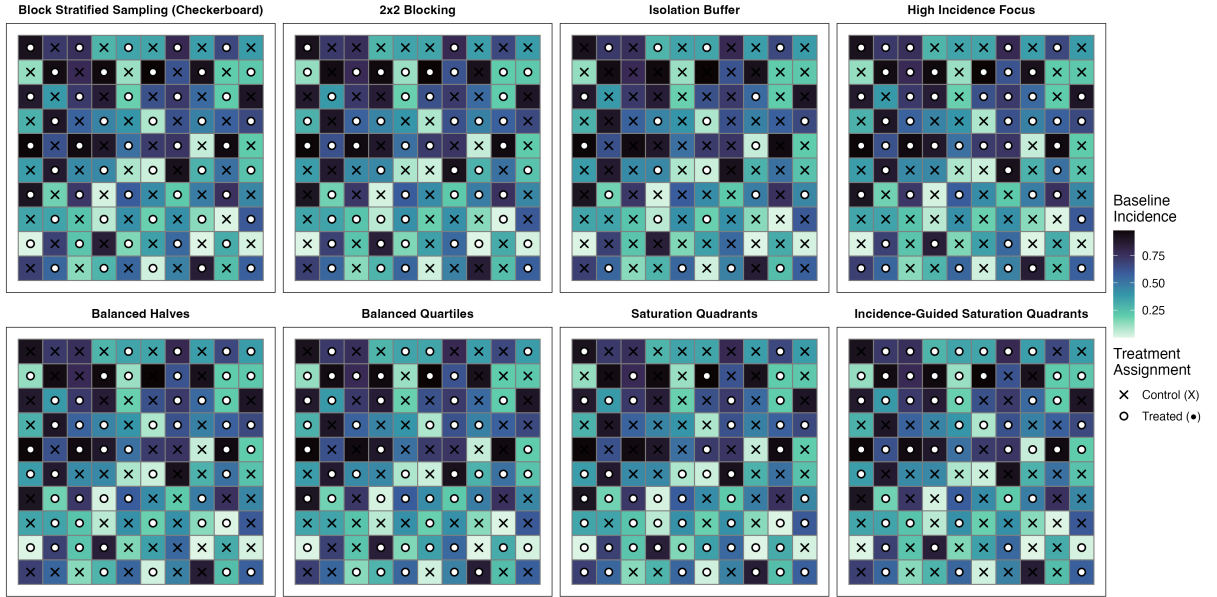


Figure 3: All 8 treatment assignment designs on a single 10×10 grid instance. Cell color represents baseline incidence (darker = higher burden). Circles (o) = treated clusters; crosses (x) = control clusters.

5.1 Design 1: Checkerboard (Strict Spatial Stratification)

$$Z_i = (x_i + y_i) \mod 2$$

Treatment in an alternating checkerboard pattern. **Deterministic**, exactly 50% treated. It is the only deterministic design: the same assignment is used for every fit, but each fit still has its own noise draw.

5.2 Design 2: High Incidence Focus

$$Z_i = \mathbf{1}[r_i > N/2], \quad r = \text{rank}(\mathbf{X}_k, \text{ties broken at random})$$

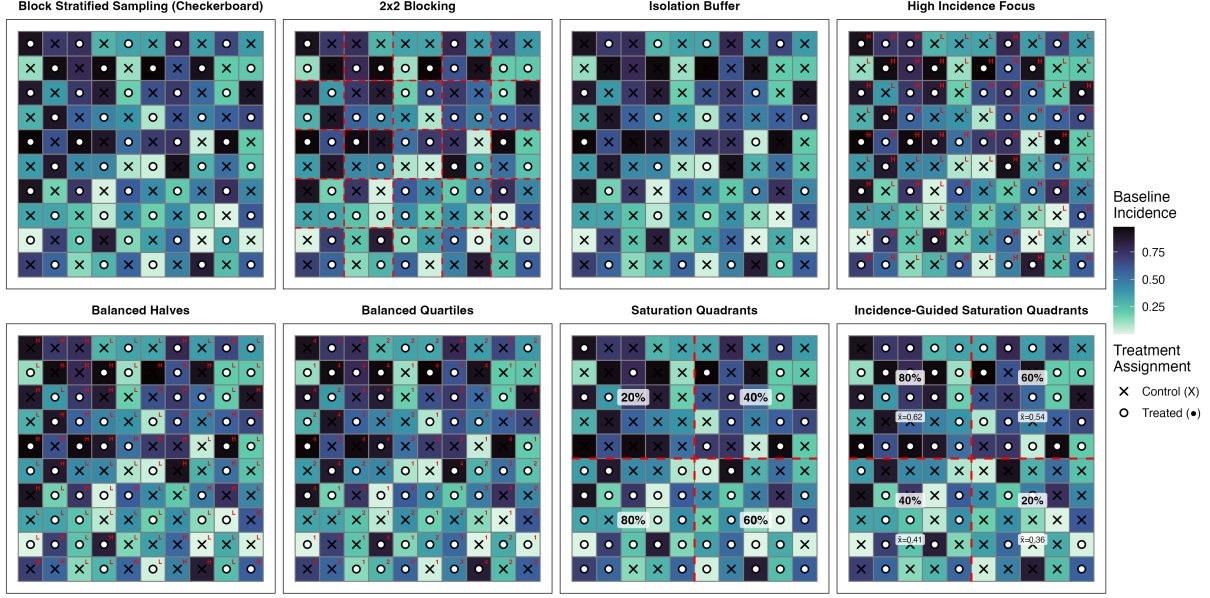


Figure 4: Same 8 designs with saturation percentage annotations. Percentage labels on the quadrant-based designs (Saturation Quadrants and Incidence-Guided Saturation Quadrants) show the within-quadrant treatment saturation rate assigned to each spatial region, illustrating how the two saturation quadrant strategies differ in how saturation levels are allocated.

Treatment assigned to the top half of clusters by incidence rank; exactly 50 treated. Ties in $\mathbf{X}_k$ are broken at random, freshly for every design draw, so the design is not deterministic.

5.3 Design 3: Saturation Quadrants

Grid partitioned into four spatial quadrants. Saturation levels $\{0.20, 0.40, 0.60, 0.80\}$ randomly assigned to quadrants. Within each quadrant, clusters randomly selected at the assigned rate.

5.4 Design 4: Isolation Buffer

Sequential treatment assignment: randomly select an available cluster, assign treatment, remove it and its neighbors from the available set. Treated clusters are never adjacent.

5.5 Design 5: 2x2 Blocking

Grid partitioned into non-overlapping 2×2 blocks. Within each block, exactly 2 of 4 clusters randomly assigned to treatment.

5.6 Design 6: Balanced Quartiles

Clusters stratified into incidence quartiles (25 each) by random-tie rank. Within each stratum 12 clusters are randomly treated, plus one more in 2 randomly chosen strata, so exactly 50 are treated.

5.7 Design 7: Balanced Halves

Clusters stratified into two halves (50 each) by random-tie incidence rank. Within each half, 25 clusters randomly assigned to treatment (exactly 50 treated). A 2-stratum analog of Balanced Quartiles.

5.8 Design 8: Incidence-Guided Saturation Quadrants

Grid partitioned into four spatial quadrants (same as Saturation Quadrants). Quadrants ranked by average baseline incidence $\bar{X}_{Q_k}$ of the design's surface $\mathbf{X}_k$ (ties broken at random) and **deterministically** assigned saturation levels:

Rank (by $\bar{X}_{Q_k}$)	Saturation p_k
1 (highest)	0.80
2	0.60
3	0.40
4 (lowest)	0.20

Within each quadrant, clusters randomly selected at the assigned rate. Unlike Saturation Quadrants, the saturation-to-quadrant mapping is guided by incidence data rather than random.

6 Estimation Methods

6.1 Difference in Means (DIM)

The naive estimator:

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_1 - \bar{Y}_0$$

DIM is biased for τ when spillover ($\gamma \neq 0$) or spatial dependence ($\rho \neq 0$) is present. DIM results exist only from the pre-revision simulation (old data-generating process, 6 designs) and are not shown in this report.

6.2 Maximum Likelihood Estimation (MLE)

The spatial autoregressive (lag) model is estimated via maximum likelihood with the project's lean engine `fit_sar_lag()`, which is the same estimator as `spatialreg::lagsarlm(method = "eigen")` (validated to agree to about 10^{-7} in $\hat{\tau}$):

$$Y_i = \rho \sum_j W_{ij} Y_j + \tau Z_i + \gamma \cdot \text{Spill}_i + \beta X_i + \varepsilon_i$$

MLE simultaneously estimates ρ and regression coefficients. The true spillover covariate $\text{Spill}(\mathbf{Z})$ is included as an **oracle regressor**. Confidence intervals for τ are 95% Wald intervals from the ML standard error. A non-oracle model ($Y \sim Z + X$, no Spill term) was also fit to the same outcomes as a sensitivity analysis; its results are not shown in this report.

Under rook contiguity the Checkerboard design gives $\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$, so the oracle model's Spill column is aliased with Z and the intercept and is dropped: τ is not identified apart from γ . These estimates are kept but flagged (`N_Aliased`, `Z_WZ_rank_deficient`); their bias is approximately $-\gamma$.

MLE is the primary estimator throughout this report. DIM serves only as a naive baseline.

7 Simulation Design

7.1 Parameter Grid

Table 2: Simulation parameter grid (2,560 total scenarios)

Parameter	Values	Levels
Incidence mode	iid, poisson, spatial	5 configs (1 iid + 2 spatial + 2 poisson)
rho_X (incidence spatial autocorr.)	N/A (iid); 0.20, 0.50 (spatial, poisson)	—
nb_type (contiguity)	queen, rook	2
rho (outcome spatial autocorr.)	0, 0.01, 0.2, 0.5	4
gamma (spillover magnitude)	0.5, 0.6, 0.7, 0.8	4
spill_type	both, control_only	2
design_id	1, 2, 3, 4, 5, 6, 7, 8	8
TOTAL		2,560

7.2 Resampling Structure

Mode	Incidence surfaces (K)	Design draws per surface (J)	Fits / scenario
MLE	10	25	250

For surface k and design draw j , $\mathbf{Z}_{kj}$ is drawn from $\mathbf{X}_k$, and each of the 250 fits gets its own noise column ε_{kj} : $\mathbf{Y}_{kj} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\tau\mathbf{Z}_{kj} + \text{Spill}(\mathbf{Z}_{kj}) + \beta\mathbf{X}_k + \varepsilon_{kj})$.

7.3 Per-Draw Reproducibility

Every random draw is preceded by `set.seed(digest::digest2int(key))`, where the key joins the fields that the draw depends on:

Draw	Key fields
$\mathbf{X}_k$ (one surface)	“X”, mode, ρ_X , k
$\mathbf{Z}_k$. (all J draws for surface k)	“Z”, mode, ρ_X , nb_type, ρ , γ , spill_type, k , design
ε (all 250 columns for a block)	“eps”, mode, ρ_X , nb_type, ρ , γ , spill_type

Each draw depends only on its key, so results are reproducible and order-independent. Within a block all 8 designs share ε and $\mathbf{X}_k$, and each τ reuses the same $\mathbf{Z}$ and ε (common random numbers).

8 Evaluation Metrics

For each scenario, let $\hat{\tau}_1, \dots, \hat{\tau}_M$ denote the M valid estimates of τ ($M = 250$ fits, excluding any with a non-finite estimate or SE).

- **Bias:** Bias = $\bar{\hat{\tau}} - \tau$
- **Standard Deviation:** SD = $\sqrt{\frac{1}{M-1} \sum_{m=1}^M (\hat{\tau}_m - \bar{\hat{\tau}})^2}$
- **Mean Squared Error:** MSE = $\frac{1}{M} \sum_{m=1}^M (\hat{\tau}_m - \tau)^2$ — **primary ranking criterion**
- **Coverage Probability:** Coverage = $\frac{1}{M} \sum_{m=1}^M \mathbf{1}[\hat{\tau}_{m,L} \leq \tau \leq \hat{\tau}_{m,U}]$
- **Power:** $\frac{1}{M} \sum_m \mathbf{1}[\hat{\tau}_{m,L} > 0]$ (one-sided)
- **Convergence Failure Rate:** Proportion of fits with a non-finite estimate or SE
- **Monte Carlo SEs:** computed from the $K = 10$ surface-level values of each metric, $\text{SE}(m) = \text{sd}(m_1, \dots, m_{10})/\sqrt{10}$, because the incidence surface is the primary sampling unit (the 250 fits are not independent)

9 Results: Overall Performance

9.1 Overall Summary

Table 3: MLE: Summary statistics across all 2,560 $\tau = 1.0$ scenarios (rook and queen pooled)

Metric	Min	Mean	Max
Bias	-0.827	-0.028	0.260
MSE	0.032	0.219	2.843
Coverage	0.000	0.890	0.990
Fail Rate	NA	0.000	NA

9.2 Design Comparison by MSE

Table 4: MLE: Design performance averaged across all $\tau = 1.0$ scenarios and incidence modes (rook and queen pooled; the Checkerboard row mixes queen with the non-identified rook case)

Design	Mean Bias	Median MSE	Mean MSE	Mean Coverage	Fail Rate
Design 3	0.018	0.065	0.080	0.938	0
Design 8	0.025	0.069	0.081	0.939	0
Design 6	0.013	0.080	0.107	0.938	0
Design 7	0.013	0.079	0.107	0.938	0
Design 4	0.003	0.146	0.147	0.939	0
Design 5	0.007	0.122	0.224	0.938	0
Design 2	0.024	0.195	0.226	0.938	0
Design 1	-0.325	0.470	0.783	0.545	0

9.3 MSE by Design and Neighbor Type

9.4 Coverage by Design

9.5 Bias-Variance Decomposition

10 Results: Formal Statistical Comparisons

The descriptive summaries above rank designs by mean MSE, but do not establish whether observed differences are *statistically significant*. Here we apply non-parametric hypothesis tests to the paired MSE values across the 320 unique parameter blocks. Each block is a unique combination of simulation parameters (Incidence Mode $\times$ Neighbor Type $\times \rho \times \gamma \times$ Spillover Type); within each block

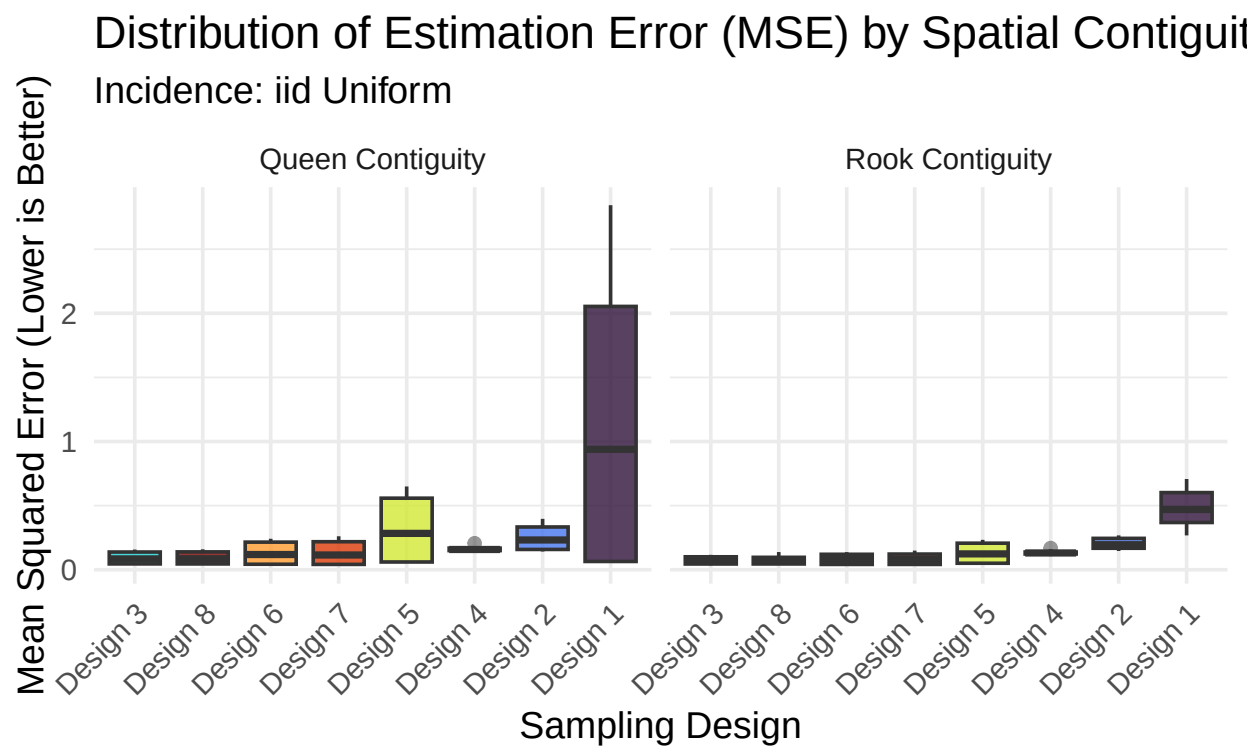


Figure 5: MLE: Distribution of MSE by design, faceted by spatial contiguity structure (iid Uniform incidence, $\tau = 1.0$).

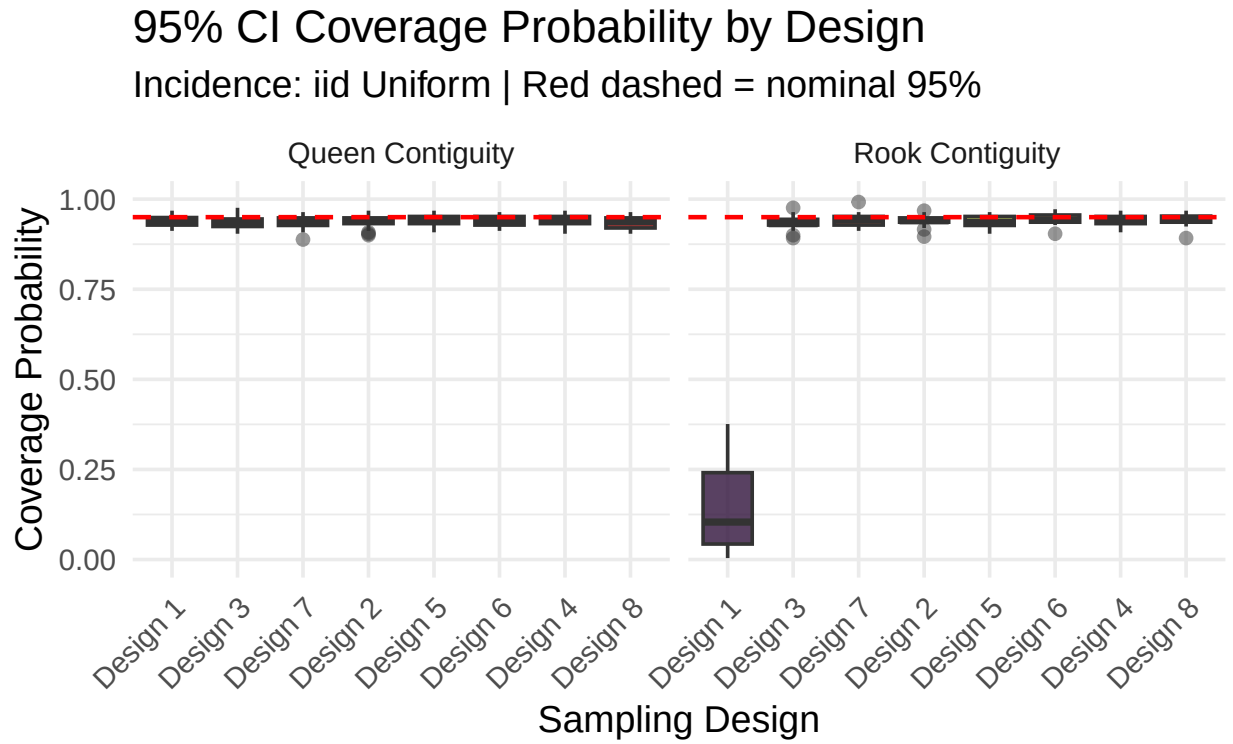


Figure 6: MLE: Coverage probability distribution by design (iid Uniform incidence, $\tau = 1.0$, rook and queen pooled). Red dashed line = nominal 95% target. The low Checkerboard values are its rook scenarios, where τ is not identified.

Bias–Variance Decomposition of MSE

Incidence: iid Uniform

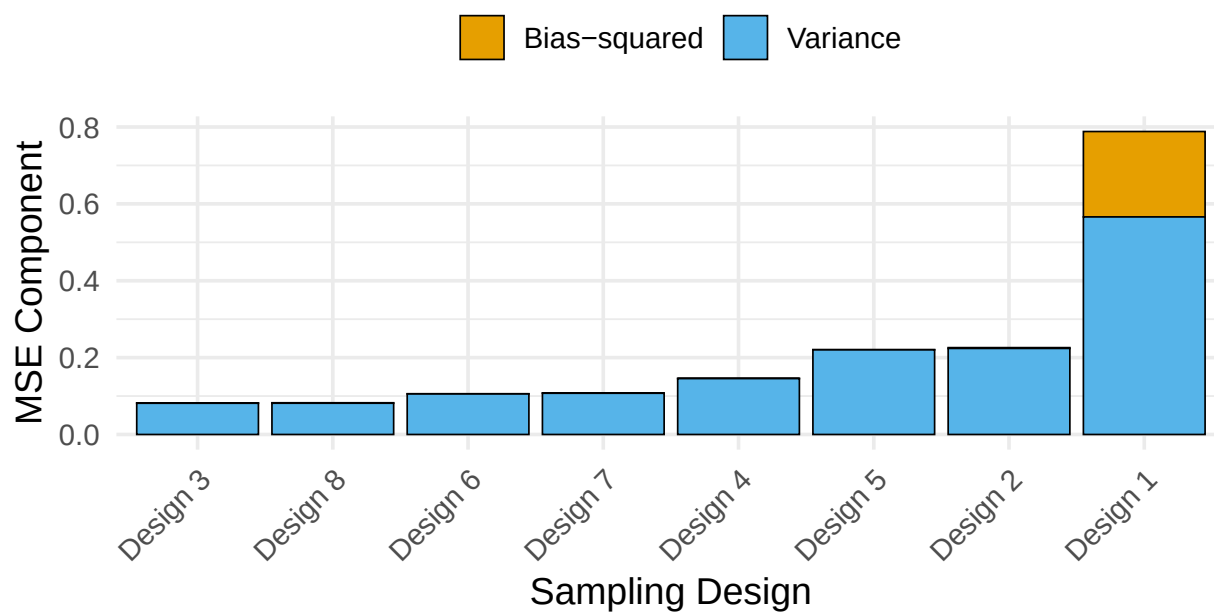


Figure 7: MLE: Bias-squared (dark) and variance (light) components of MSE per design for iid Uniform incidence ($\tau = 1.0$, rook and queen pooled).

all 8 designs are evaluated under identical conditions, creating a paired repeated-measures structure. **These tests pool rook and queen blocks at $\tau = 1.0$.** For the primary queen structure alone (160 blocks), the Friedman statistic is $\chi^2 = 711.29$ ($df = 7$, $p < 2 \times 10^{-16}$); queen, rook and pooled results for all 8 designs are in `results/eight_design_supplementary/eight_design_summary.txt`. Blocks within an incidence configuration share the same 10 incidence surfaces, so treating blocks as independent overstates the information about surface-to-surface variation.

Three complementary tests are used: (1) the **Friedman test** (omnibus — is there any significant difference among designs?), (2) **Nemenyi post-hoc** comparisons with family-wise error control (which specific pairs differ?), and (3) pairwise **Wilcoxon signed-rank tests** with Holm correction (sensitivity check). Significance level $\alpha = 0.05$ throughout.

10.1 Friedman Omnibus Test

Table 5: Friedman omnibus test: are any design MSE ranks significantly different?

Chi-squared	df	P-value	N Blocks	Decision
1573.47	7	<2e-16	320	Reject H0

The pooled Friedman test ($\chi^2 = 1573.47$, $p < 2.2 \times 10^{-16}$) decisively rejects the null hypothesis that all 8 designs perform equally. The table below shows each design’s **average MSE rank** across all blocks (rank 1 = best):

Table 6: Average MSE rank per design across all 320 blocks (rook and queen pooled; 1 = best).

Design	Average Rank
Design 3	2.191
Design 8	2.328
Design 6	3.138
Design 7	3.175
Design 4	5.181
Design 5	5.759
Design 2	6.866
Design 1	7.362

10.2 Critical Difference Diagram

How to read this diagram: Designs positioned closer together are more similar in performance. A thick bar connecting two designs means their difference is *not* statistically significant. The red arrow labelled “CD” shows the critical difference threshold — any two designs separated by more than this distance are significantly different.

The headline finding: **Saturation Quadrants and Incidence-Guided Saturation Quadrants form the top equivalence group** — statistically indistinguishable from each other (Nemenyi $p = 0.997$ pooled, 0.985 queen) and significantly better than every other design in both the pooled

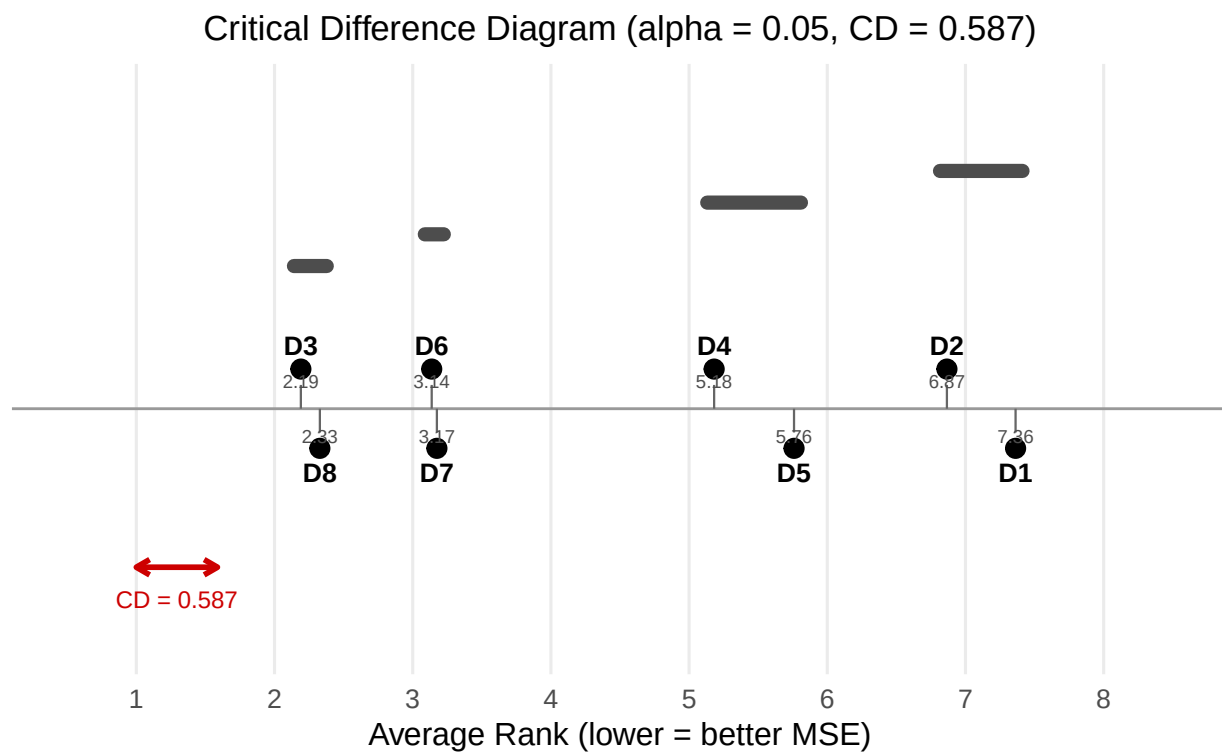


Figure 8: Critical Difference (CD) diagram from Nemenyi post-hoc analysis. Designs are placed on a number line by average MSE rank (lower = better). Thick bars connect designs that are NOT significantly different at alpha = 0.05. The red arrow shows the CD threshold.

and the queen-only analysis. Under rook alone they are not significantly separated from Balanced Quartiles or Balanced Halves (Nemenyi p from 0.10 to 0.33). **Checkerboard has the worst pooled average rank, but it is not significantly worse than every alternative:** pooled, it does not differ significantly from High Incidence Focus (Nemenyi $p = 0.17$); under queen, Checkerboard, High Incidence Focus and 2x2 Blocking are not significantly different from one another (Nemenyi $p = 1.00, 0.62, 0.55$). Under rook, Checkerboard ranks last in nearly every block (average rank 7.99 of 8) because τ is not identified there.

10.3 Pairwise Significance: MSE Distributions

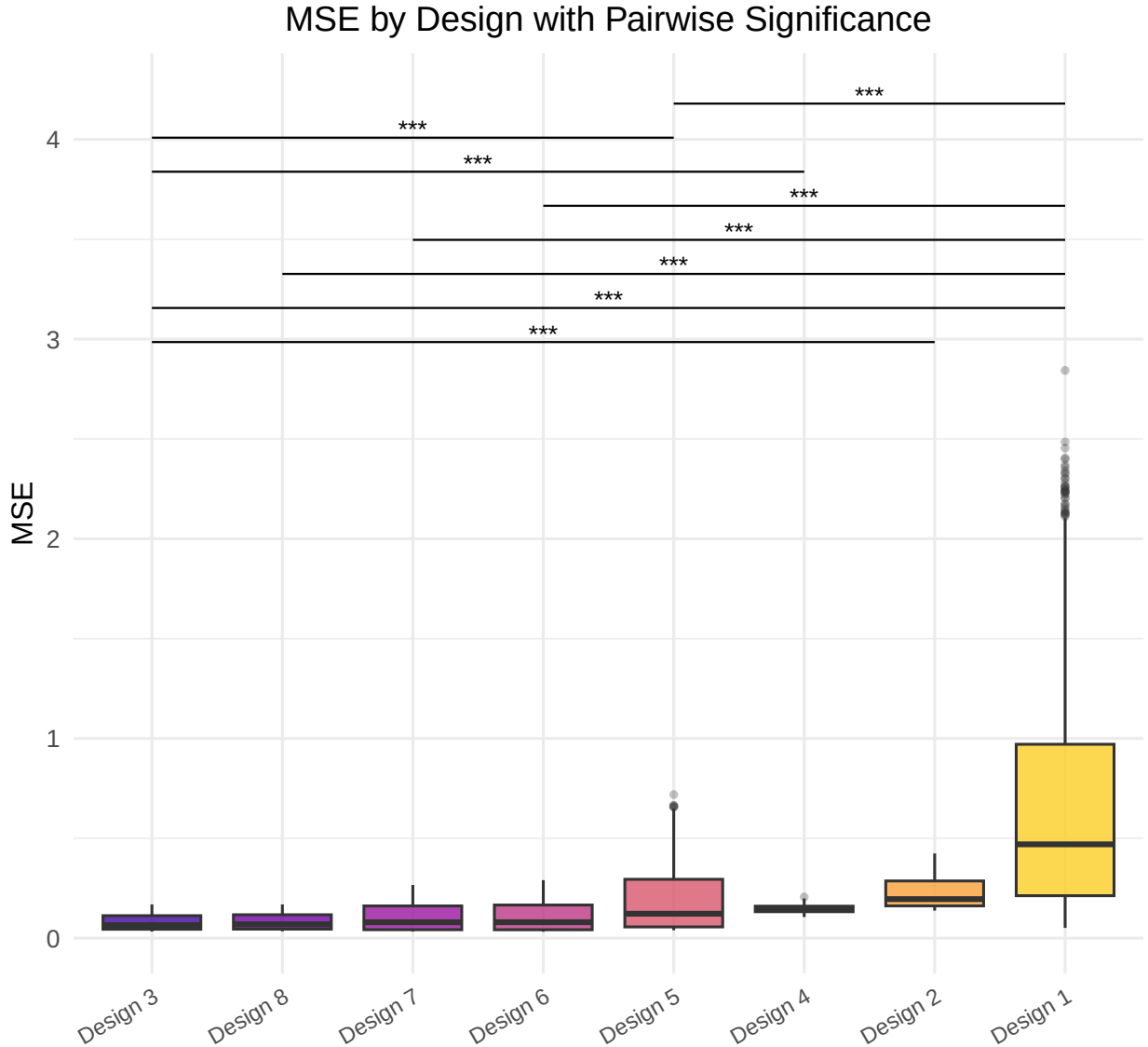


Figure 9: MSE distribution by design ($\tau = 1.0$, rook and queen pooled; ordered by median MSE, best on left). Significance brackets show pairwise Wilcoxon signed-rank test results (Holm-adjusted). Stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

The boxplot (rook and queen pooled) confirms the rank-based findings: Saturation Quadrants and Incidence-Guided Saturation Quadrants have the lowest and most compact MSE distributions, while Checkerboard has both the highest median and the widest spread. Its MSE comes from two sources: a large variance under queen with control-only spillover (bias near 0), where the Spill covariate is nearly collinear with Z , and, under rook, a bias of about $-\gamma$ because τ is not identified. Significance brackets annotate the most informative pairwise comparisons.

10.4 Pairwise P-value Heatmap



Figure 10: Pairwise Wilcoxon signed-rank p-values (Holm-adjusted; rook and queen pooled). Red = strongly significant ($p < 0.001$), orange = significant ($p < 0.01$ or < 0.05), grey/green = not significant.

Of the 28 pairwise comparisons, **24 are statistically significant** under Wilcoxon (Holm) and **24**

under Nemenyi post-hoc. The 4 non-significant pair(s) are:

Table 7: Design pairs with no statistically significant MSE difference.

Design A	Design B	Wilcoxon p	Nemenyi p
Design 2	Design 5	0.1241	0.0000
Design 3	Design 8	0.2976	0.9967
Design 4	Design 5	0.1527	0.0570
Design 6	Design 7	0.6513	1.0000

10.5 Robustness: Conditional Tests by Parameter

To confirm that the design rankings are not an artifact of any particular parameter setting, we repeat the Friedman test within strata defined by each simulation parameter individually.

Table 8: Friedman test within each parameter stratum. ‘Best Group’ lists designs statistically indistinguishable from the best-ranked design.

Parameter	Stratum	Chi-sq	P-value	N Blocks	Best Group
Rho	Rho = 0	394.65	$< 2.2\text{e-}16$	80	D3, D8, D6, D7
Rho	Rho = 0.01	403.31	$< 2.2\text{e-}16$	80	D8, D3, D7, D6
Rho	Rho = 0.2	398.61	$< 2.2\text{e-}16$	80	D3, D8, D7
Rho	Rho = 0.5	380.75	$< 2.2\text{e-}16$	80	D3, D8, D6, D7
Gamma	Gamma = 0.5	396.29	$< 2.2\text{e-}16$	80	D8, D3, D6, D7
Gamma	Gamma = 0.6	389.30	$< 2.2\text{e-}16$	80	D3, D8, D7
Gamma	Gamma = 0.7	390.98	$< 2.2\text{e-}16$	80	D3, D8, D6, D7
Gamma	Gamma = 0.8	399.49	$< 2.2\text{e-}16$	80	D8, D3, D6, D7
Spillover_Type	Spillover_Type = both	889.37	$< 2.2\text{e-}16$	160	D6, D7, D3
Spillover_Type	Spillover_Type = control_only	1019.96	$< 2.2\text{e-}16$	160	D8, D3
Neighbor_Type	Neighbor_Type = queen	711.29	$< 2.2\text{e-}16$	160	D3, D8
Neighbor_Type	Neighbor_Type = rook	900.22	$< 2.2\text{e-}16$	160	D3, D8, D6, D7
Incidence_Mode	Incidence_Mode = iid	314.23	$< 2.2\text{e-}16$	64	D3, D8, D6, D7
Incidence_Mode	Incidence_Mode = poisson	633.48	$< 2.2\text{e-}16$	128	D3, D8
Incidence_Mode	Incidence_Mode = spatial	626.54	$< 2.2\text{e-}16$	128	D3, D8

The Friedman test is **significant in every stratum** ($p < 2.2 \times 10^{-16}$ in all cases). Saturation Quadrants appears in the top equivalence group in every stratum, and Incidence-Guided Saturation Quadrants in every stratum except bidirectional (“both”) spillover, where Balanced Quartiles ranks first and Balanced Halves and Saturation Quadrants are statistically tied with it. Except for the Neighbor_Type rows, each stratum pools rook and queen blocks.

For full methods background (Friedman test derivation, CD formula, Wilcoxon procedure with references) and complete pairwise p-value matrices, see the companion report: **Statistical Comparison of Treatment Assignment Designs** (11_statistical_comparisons_report.pdf in the results directory).

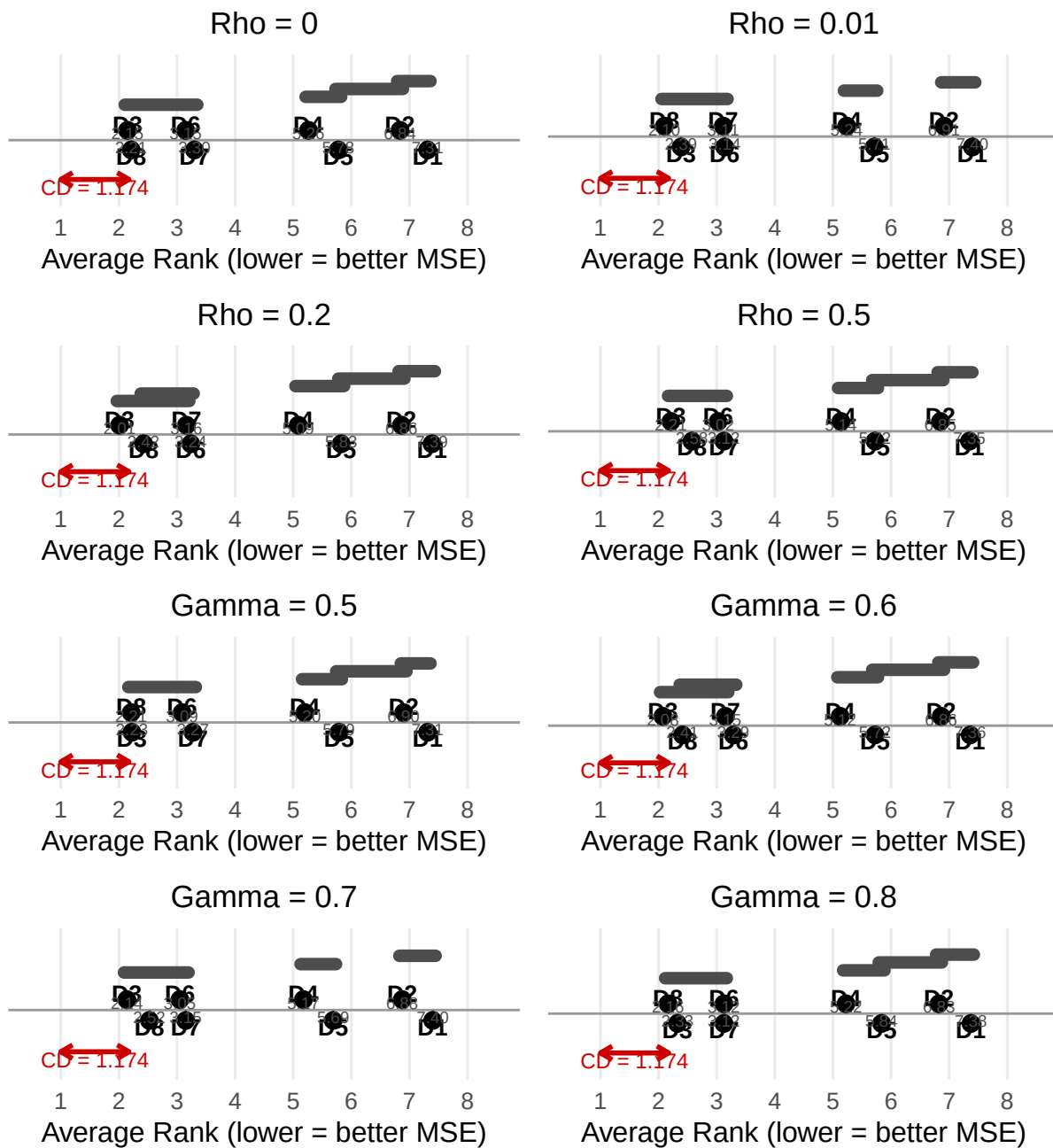


Figure 11: CD diagrams by outcome spatial autocorrelation (Rho, top) and spillover magnitude (Gamma, bottom). Rankings are consistent across levels.

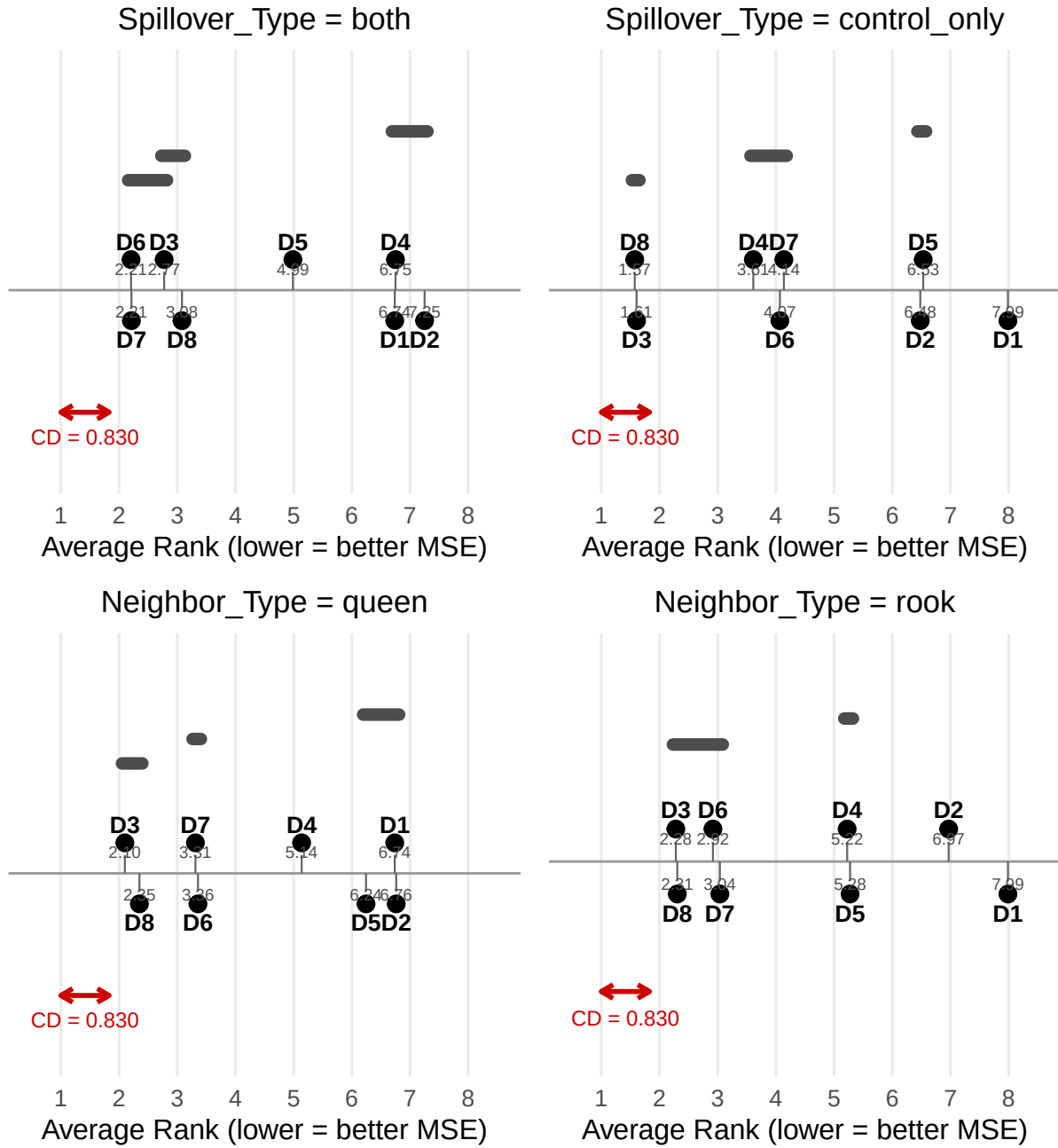


Figure 12: CD diagrams by spillover type (top) and neighbor contiguity type (bottom). The top of the ranking is similar across spillover types and contiguity definitions; the bottom differs (under rook, Checkerboard is last because tau is not identified).

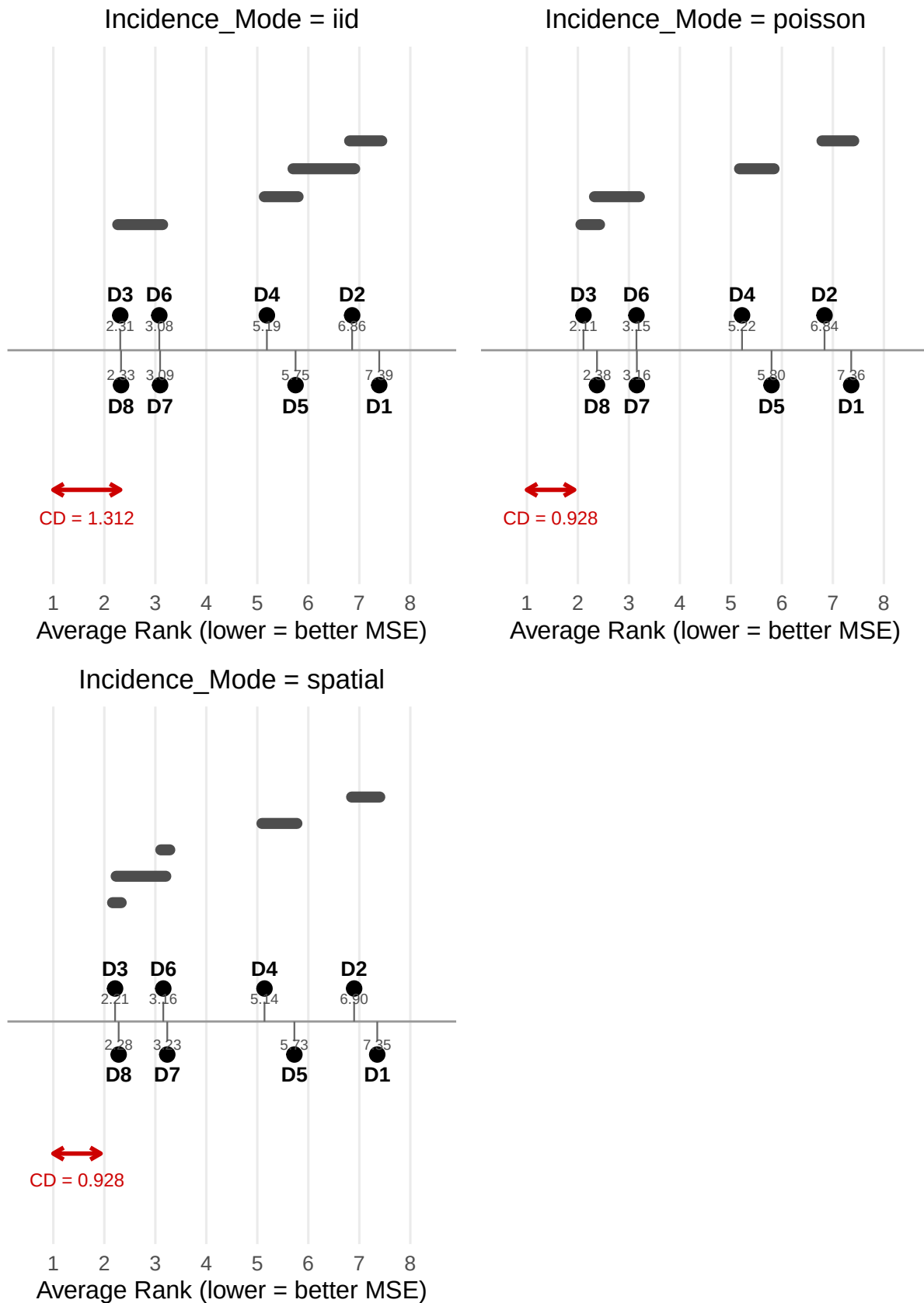


Figure 13: CD diagrams by incidence generation mode. Design rankings hold across all five incidence configurations.

10.6 Summary of Statistical Findings

Taken together, the Friedman test, Nemenyi post-hoc comparisons, and pairwise Wilcoxon signed-rank tests (all pooling rook and queen at $\tau = 1.0$ unless stated) converge on a clear and consistent picture:

- **Design differences are real, not chance.** The pooled Friedman omnibus test ($\chi^2 = 1573.47$, $p < 2.2 \times 10^{-16}$) provides overwhelming evidence that the 8 designs do not perform equally. The average rank gap between best and worst is 5.17 points on an 8-point scale.
- **Saturation Quadrants and Incidence-Guided Saturation Quadrants are the recommended designs.** Saturation Quadrants (Design 3, avg rank 2.19) and Incidence-Guided Saturation Quadrants (Design 8, avg rank 2.33) form the top equivalence group — they are statistically indistinguishable from each other and, pooled and under queen, significantly better than every other design (under rook alone, not significantly separated from Balanced Quartiles or Balanced Halves). The data do not favor one of the two over the other.
- **Avoid Checkerboard.** With a pooled average rank of 7.36, Checkerboard is last. Under rook, τ is not identified (bias about $-\gamma$, coverage 0.15); under queen, its coverage is near nominal but its MSE is the largest of the 8 designs (1.087 vs 0.090 for Saturation Quadrants), and it is not significantly different from High Incidence Focus or 2x2 Blocking.
- **24 of 28 design pairs are significantly different** (pooled Nemenyi post-hoc; 24 under Wilcoxon/Holm). The non-significant pairs include the two near-identical pairs (Saturation Quadrants vs Incidence-Guided Saturation Quadrants; Balanced Quartiles vs Balanced Halves) and pairs among the lower-ranked designs (see the table above).
- **Rankings are robust.** The Friedman test is significant in every parameter stratum tested ($p < 2.2 \times 10^{-16}$ across all levels of ρ , γ , spillover type, neighbor type, and incidence mode). Saturation Quadrants is in the top equivalence group in every stratum, and Incidence-Guided Saturation Quadrants in every stratum except bidirectional spillover, so the recommendation generalises beyond any single scenario. Per-stratum tests share incidence surfaces and noise draws, so they are not independent confirmations of one another.

11 Results: Design Rankings by Incidence Mode (Q1)

11.1 Cross-Config MSE Rankings

11.2 Cross-Config Coverage Rankings

11.3 Detailed Rankings by Incidence Config

Table 9: MLE design rankings for iid Uniform ($\tau = 1.0$, rook and queen pooled)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
------	--------	---------	--------------	----------

1	D3: Sat. Quadrants	0.0818	0.935	0.0186
2	D8: Inc.-Guided Sat. Q.	0.0822	0.940	0.0246
3	D6: Balanced Quartiles	0.1056	0.942	0.0079
4	D7: Balanced Halves	0.1075	0.938	0.0135
5	D4: Isolation Buffer	0.1455	0.941	0.0081
6	D5: 2x2 Blocking	0.2200	0.939	0.0038
7	D2: High Inc. Focus	0.2251	0.940	0.0265
8	D1: Checkerboard	0.7860	0.545	-0.3260

Table 10: MLE design rankings for Poisson ($\rho X = 0.20$) ($\tau = 1.0$, rook and queen pooled)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	D3: Sat. Quadrants	0.0776	0.942	0.0174
2	D8: Inc.-Guided Sat. Q.	0.0814	0.938	0.0267
3	D6: Balanced Quartiles	0.1065	0.937	0.0144
4	D7: Balanced Halves	0.1077	0.937	0.0115
5	D4: Isolation Buffer	0.1491	0.937	0.0015
6	D2: High Inc. Focus	0.2172	0.941	0.0278
7	D5: 2x2 Blocking	0.2280	0.939	0.0080
8	D1: Checkerboard	0.7963	0.542	-0.3232

Table 11: MLE design rankings for Poisson ($\rho X = 0.50$) ($\tau = 1.0$, rook and queen pooled)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	D3: Sat. Quadrants	0.0804	0.939	0.0164
2	D8: Inc.-Guided Sat. Q.	0.0806	0.940	0.0242
3	D7: Balanced Halves	0.1066	0.937	0.0104
4	D6: Balanced Quartiles	0.1101	0.935	0.0135
5	D4: Isolation Buffer	0.1475	0.939	0.0026
6	D2: High Inc. Focus	0.2244	0.936	0.0245
7	D5: 2x2 Blocking	0.2244	0.936	0.0070
8	D1: Checkerboard	0.7797	0.544	-0.3263

Table 12: MLE design rankings for Spatial ($\rho X = 0.20$) ($\tau = 1.0$, rook and queen pooled)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	D8: Inc.-Guided Sat. Q.	0.0795	0.942	0.0254
2	D3: Sat. Quadrants	0.0800	0.939	0.0164
3	D6: Balanced Quartiles	0.1071	0.937	0.0137
4	D7: Balanced Halves	0.1085	0.936	0.0171
5	D4: Isolation Buffer	0.1449	0.941	0.0019
6	D5: 2x2 Blocking	0.2201	0.938	0.0108
7	D2: High Inc. Focus	0.2399	0.937	0.0191
8	D1: Checkerboard	0.7749	0.547	-0.3228

Design Recommendation: Average MSE by Incidence Mode

Rank labels shown above bars (1 = best). MLE estimator.

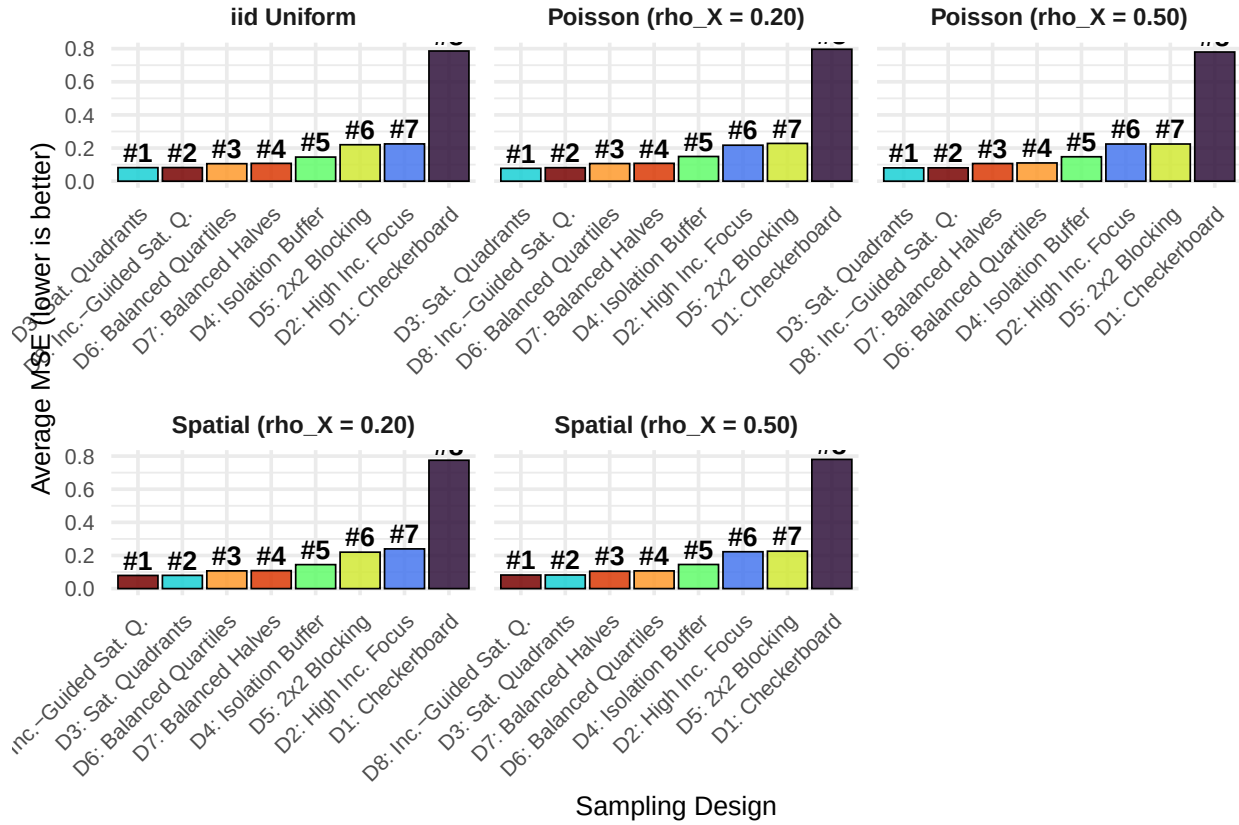


Figure 14: Average MSE per design for each incidence configuration ($\tau = 1.0$, rook and queen pooled). Bars sorted left-to-right (lowest = best). Rank labels above bars.

Design Recommendation: Average Coverage by Incidence Mode

Red dashed = nominal 95%. MLE estimator.

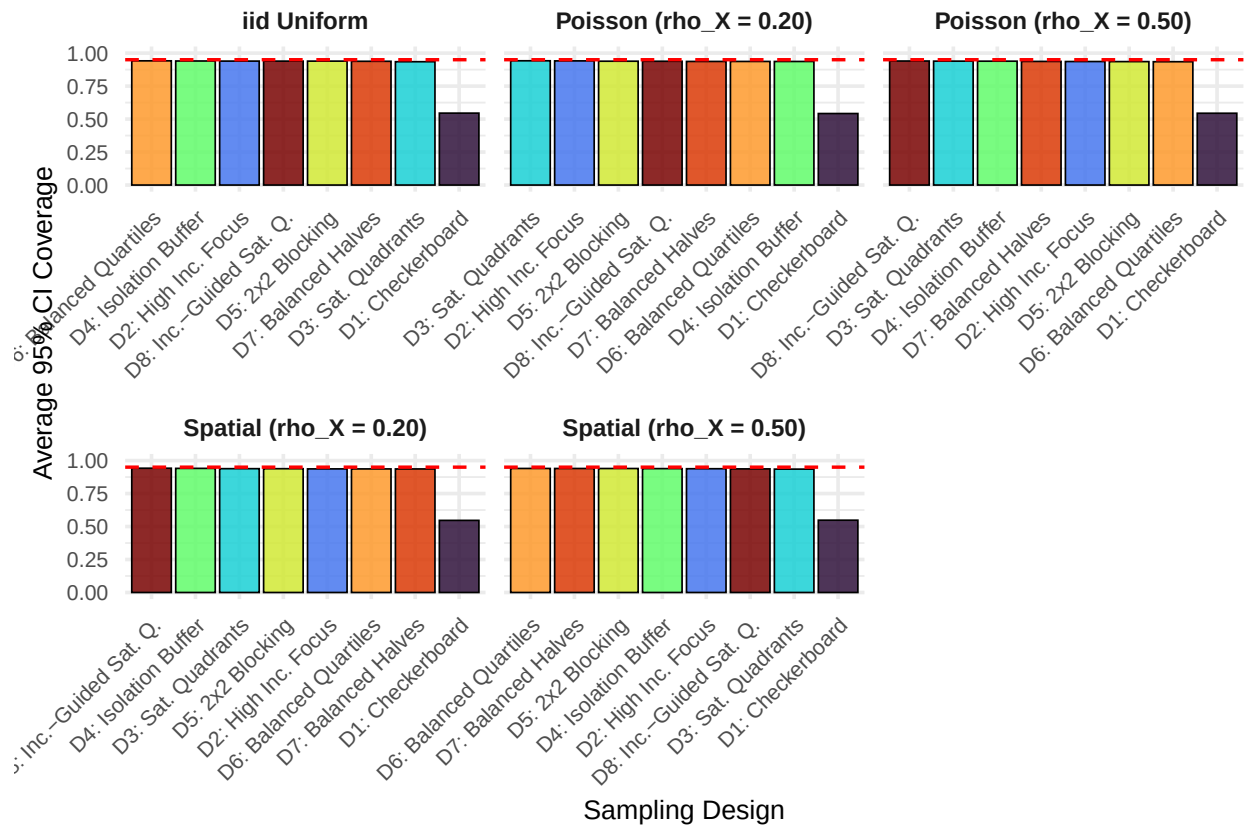


Figure 15: Average 95% CI coverage probability per design for each incidence configuration ($\tau = 1.0$, rook and queen pooled). Red dashed line = nominal 0.95. Checkerboard's low pooled coverage comes entirely from rook, where τ is not identified; under queen it is near nominal.

Table 13: MLE design rankings for Spatial ($\rho_X = 0.50$) ($\tau = 1.0$, rook and queen pooled)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	D8: Inc.-Guided Sat. Q.	0.0821	0.937	0.0256
2	D3: Sat. Quadrants	0.0822	0.935	0.0206
3	D7: Balanced Halves	0.1050	0.940	0.0123
4	D6: Balanced Quartiles	0.1071	0.940	0.0148
5	D4: Isolation Buffer	0.1456	0.939	-0.0012
6	D2: High Inc. Focus	0.2227	0.938	0.0246
7	D5: 2x2 Blocking	0.2258	0.940	0.0043
8	D1: Checkerboard	0.7801	0.548	-0.3286

12 Results: Parameter Sensitivity (Q2)

12.1 Effect of Outcome Spatial Autocorrelation (Rho)

12.2 Effect of Spillover Magnitude (Gamma)

12.3 Sensitivity Analysis (Eta-squared)

Table 14: MLE: Eta-squared (proportion of MSE variance explained) by parameter, for iid Uniform incidence ($\tau = 1.0$, rook and queen pooled).

Design	Rho	Gamma	Spillover_Type	Neighbor_Type
Design 1	0.0009	0.0142	0.4125	0.1445
Design 2	0.0011	0.0134	0.7294	0.0832
Design 3	0.0042	0.0002	0.8537	0.0552
Design 4	0.0364	0.0052	0.0014	0.5209
Design 5	0.0010	0.0012	0.6207	0.1933
Design 6	0.0009	0.0003	0.7696	0.1097
Design 7	0.0016	0.0011	0.7634	0.1031
Design 8	0.0016	0.0044	0.8334	0.0645

12.4 MSE Per Design Across Gamma

\$`Design 1`

\$`Design 2`

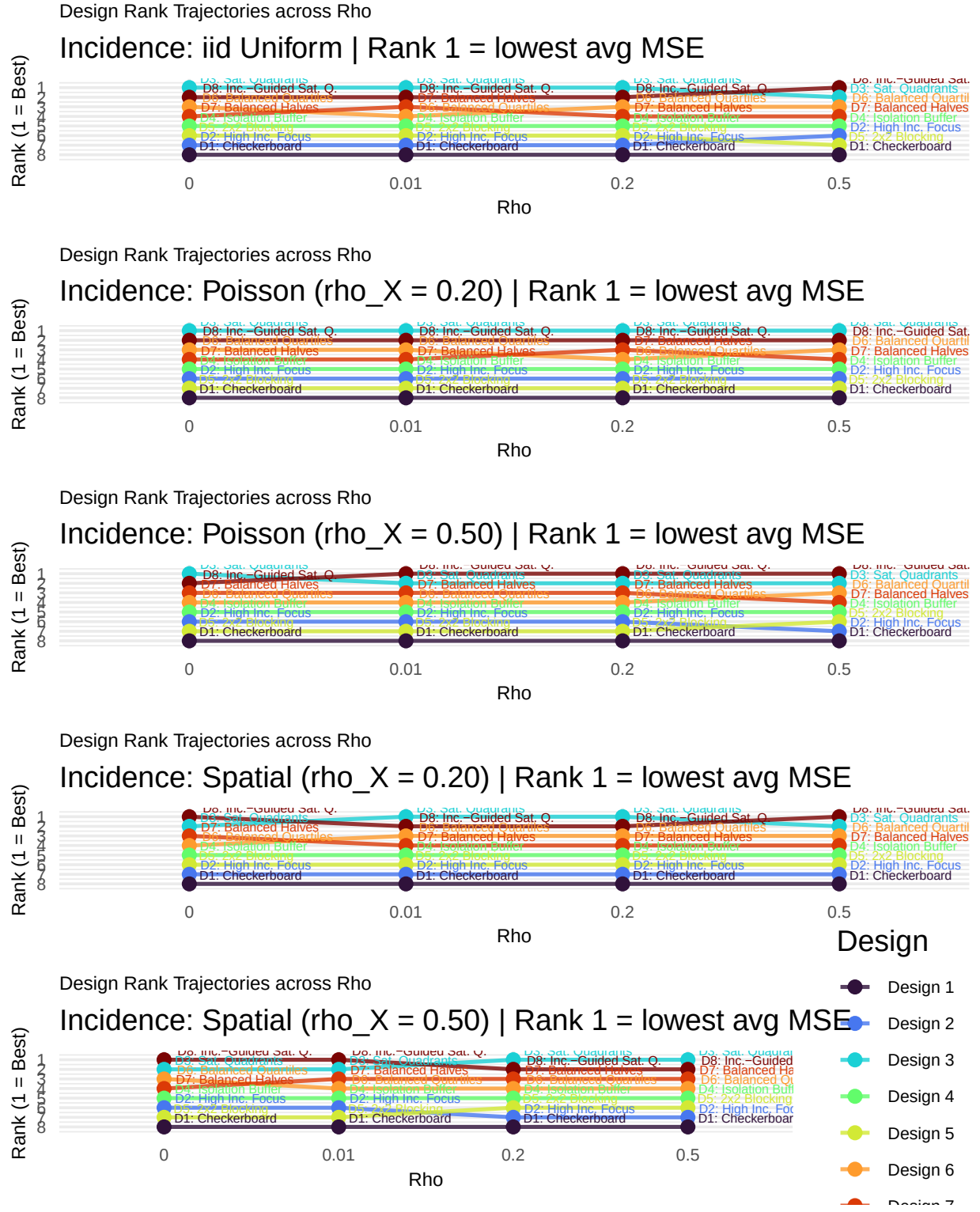


Figure 16: Design rank trajectories across Rho for each incidence config (tau = 1.0, rook and queen pooled). Rank 1 = best. Flat lines indicate robust designs; crossing lines reveal rank reversals.

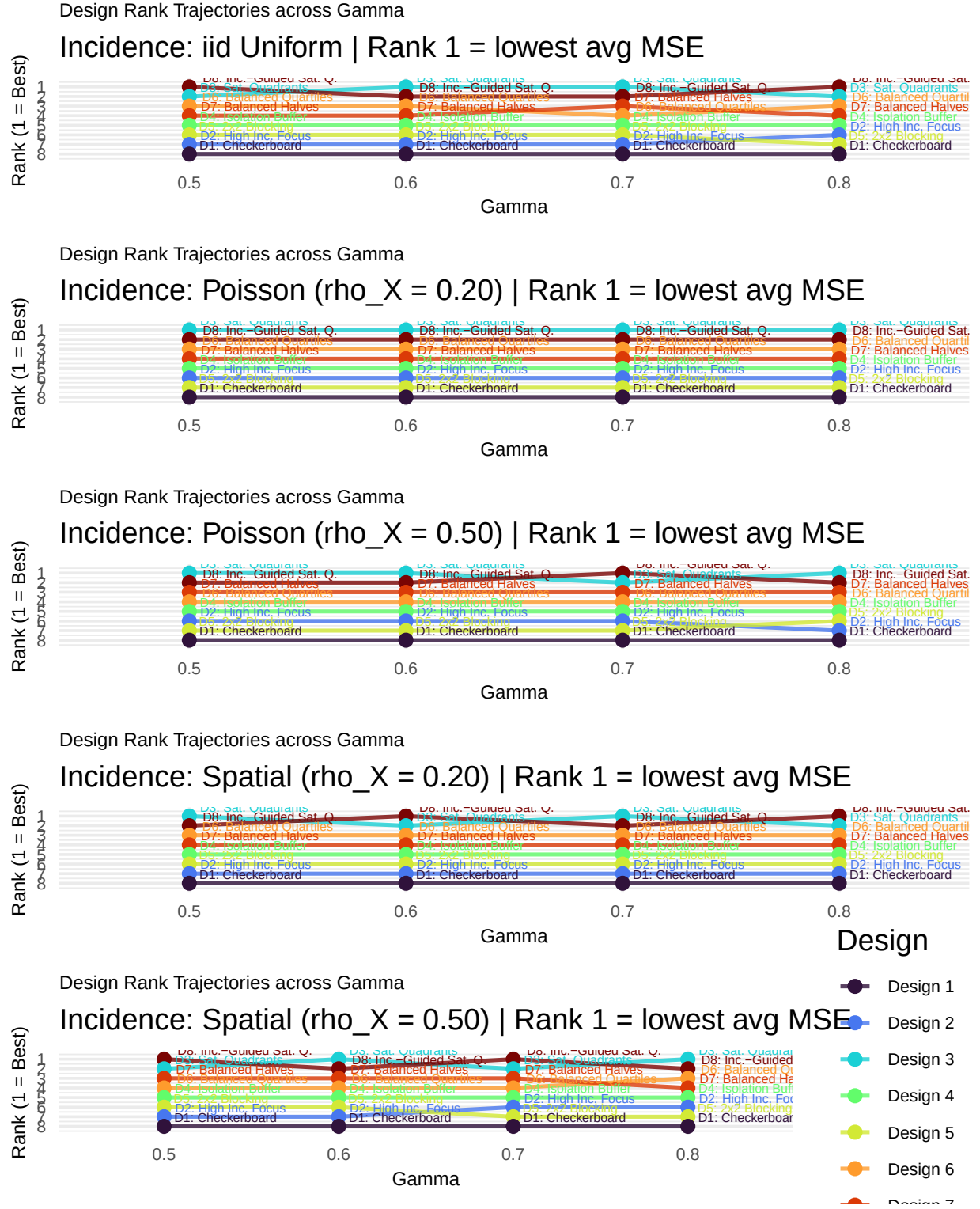


Figure 17: Design rank trajectories across Gamma (tau = 1.0, rook and queen pooled). Gamma has minimal effect on design rankings under MLE: all trajectory lines are nearly flat.

MSE Dynamics for Design 1

Incidence: iid Uniform

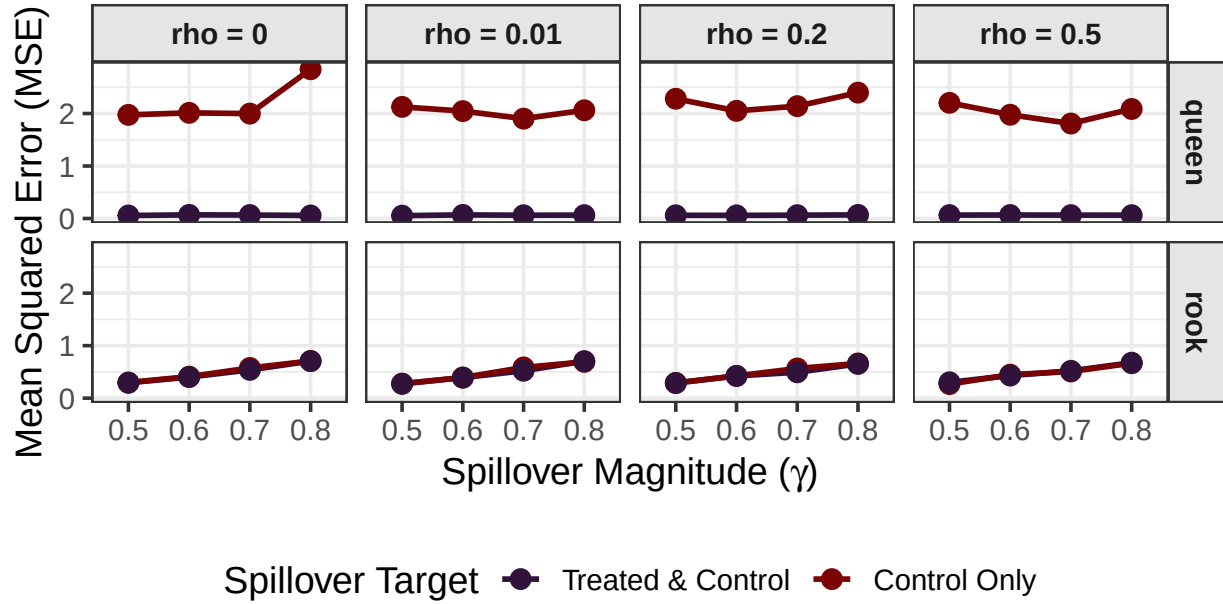


Figure 18: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 2

Incidence: iid Uniform

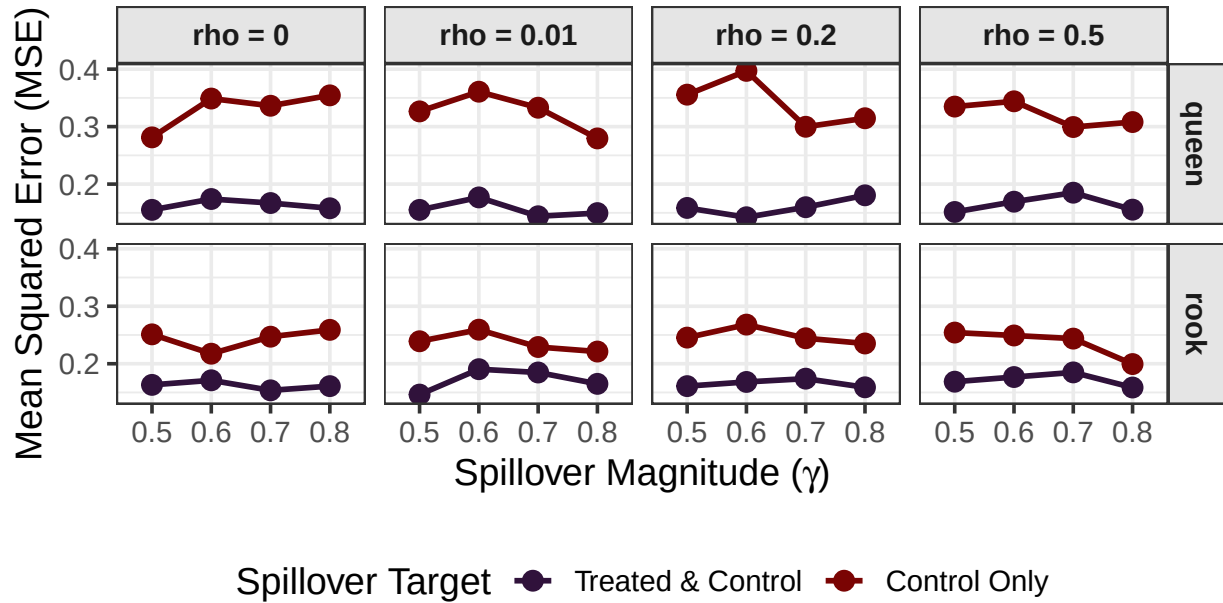


Figure 19: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 3

Incidence: iid Uniform

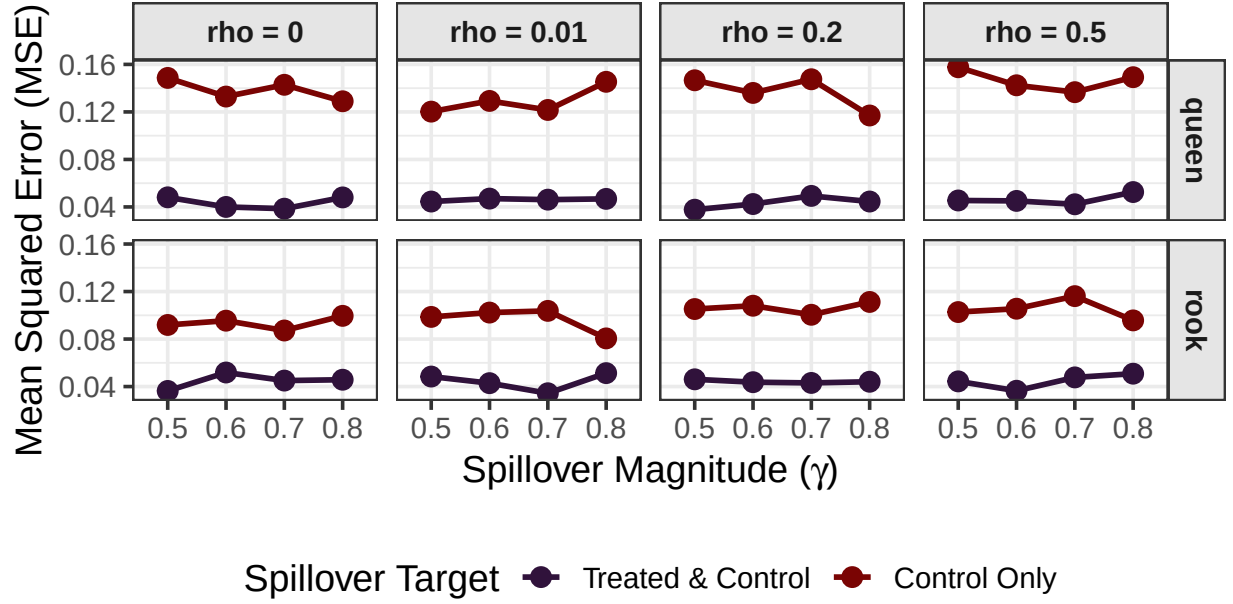


Figure 20: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 4

Incidence: iid Uniform

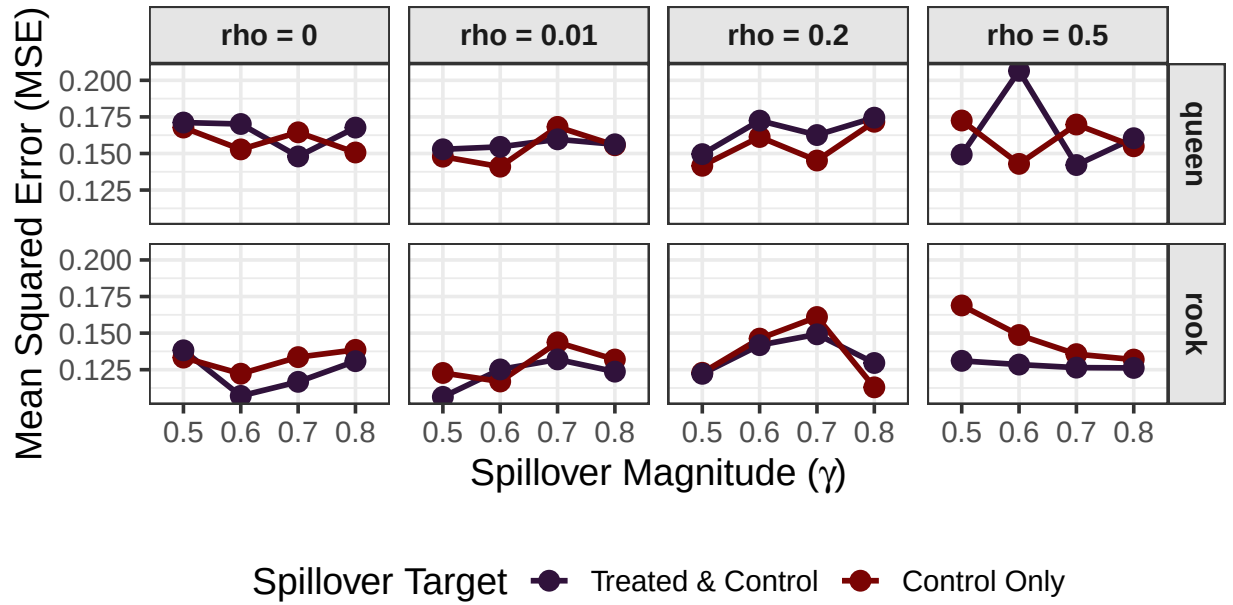


Figure 21: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 5

Incidence: iid Uniform

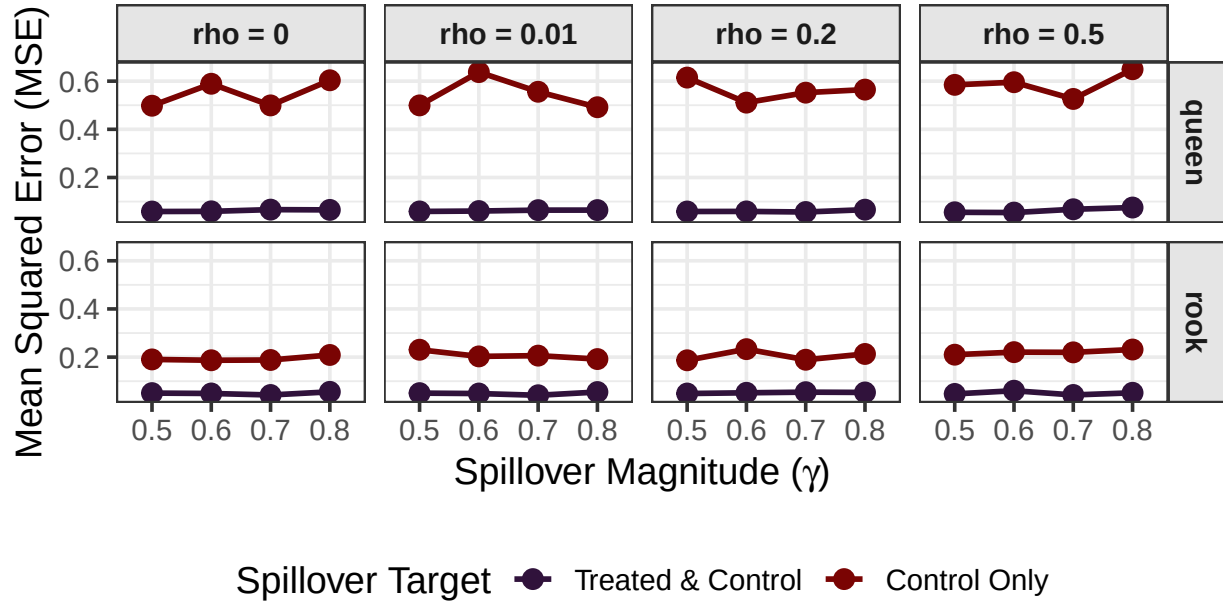


Figure 22: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 6

Incidence: iid Uniform

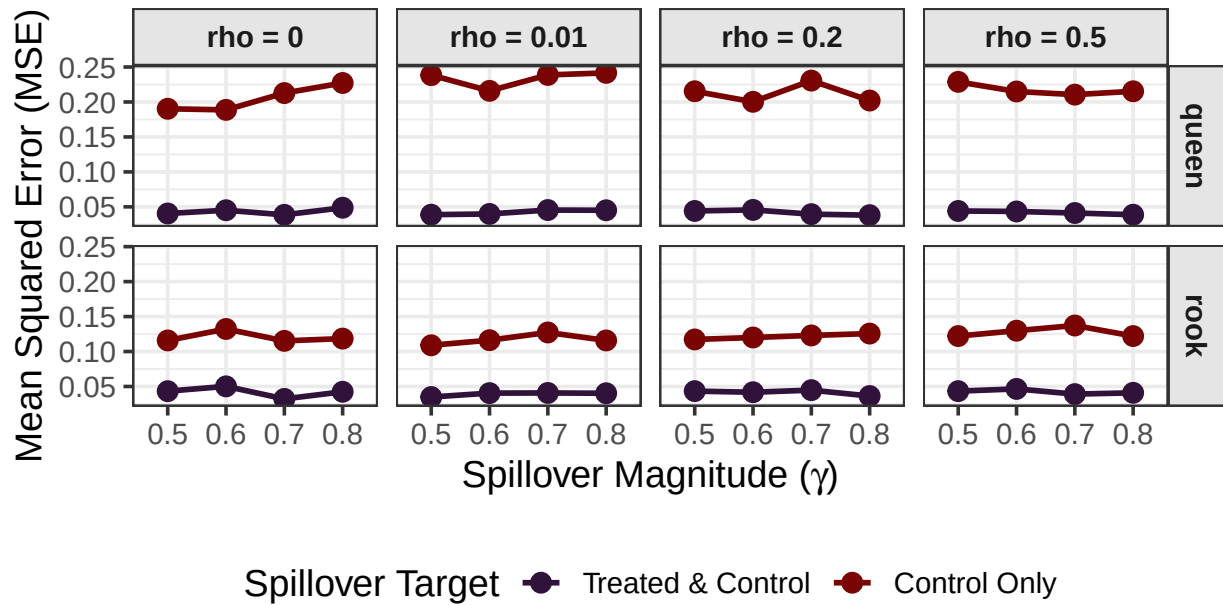


Figure 23: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 7

Incidence: iid Uniform

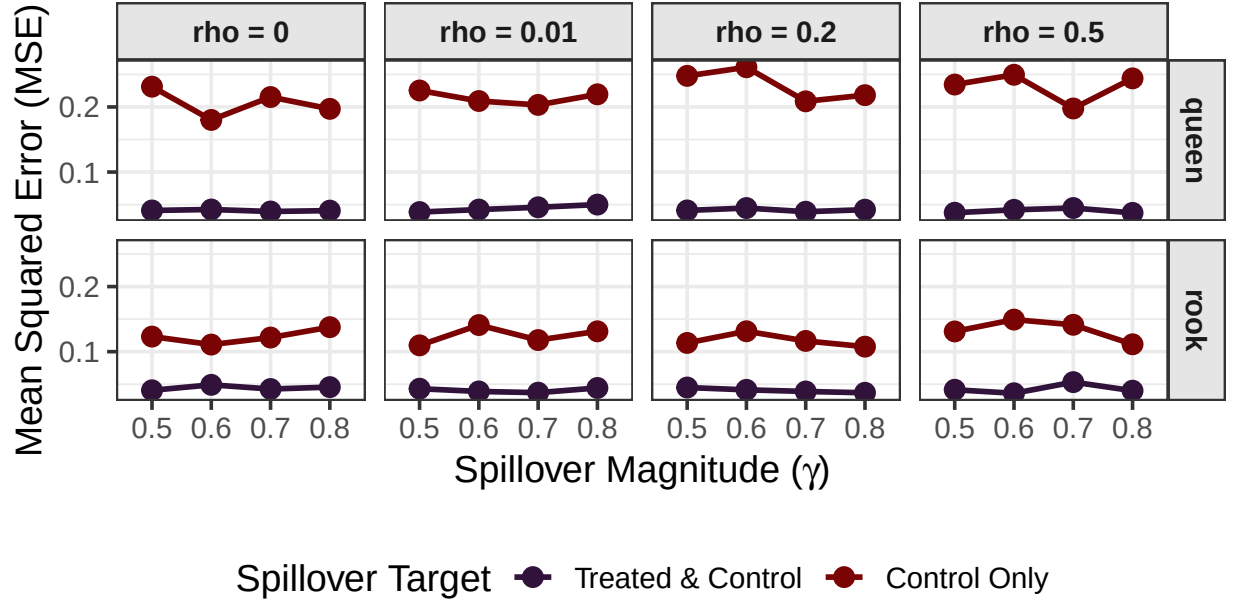


Figure 24: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 8

Incidence: iid Uniform

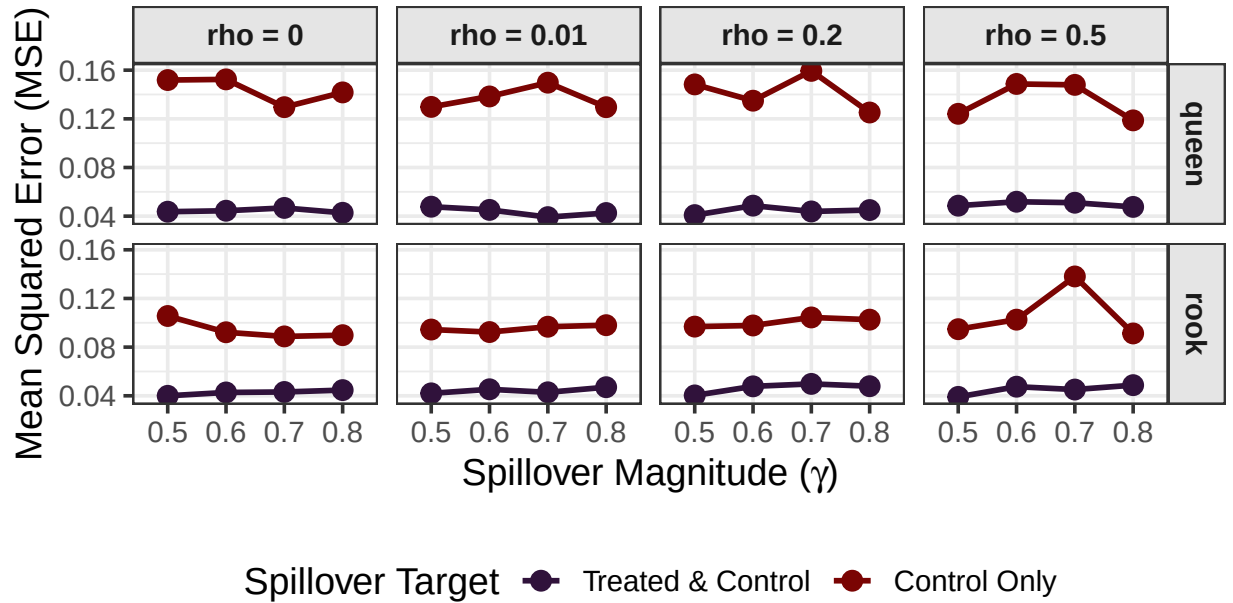


Figure 25: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 1

Incidence: iid Uniform

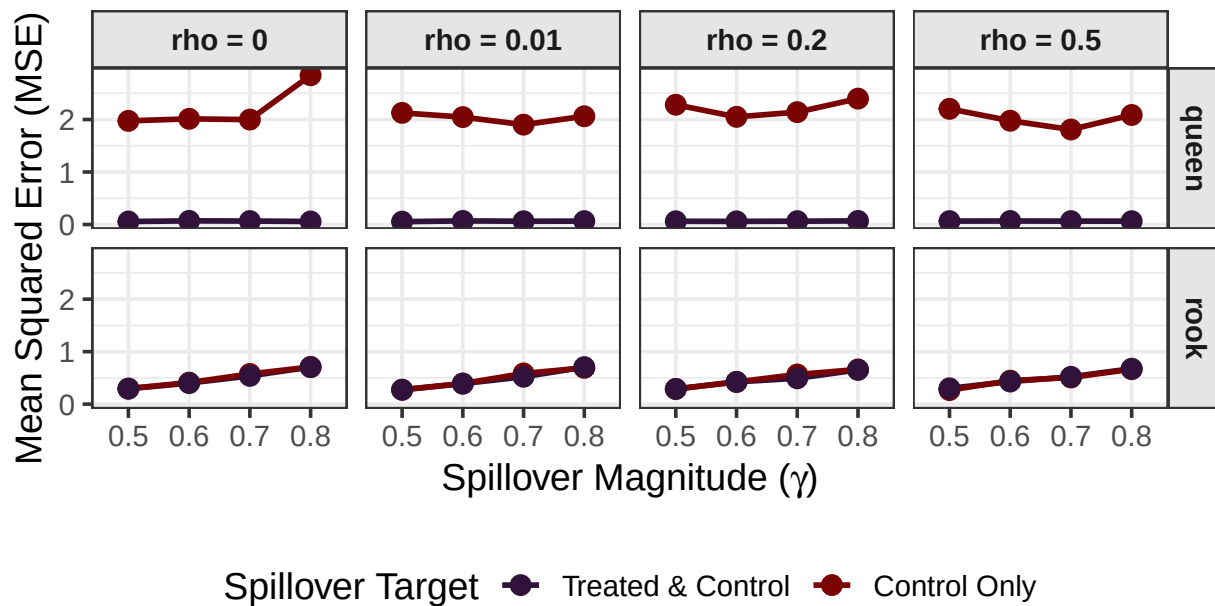


Figure 26: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 2

Incidence: iid Uniform

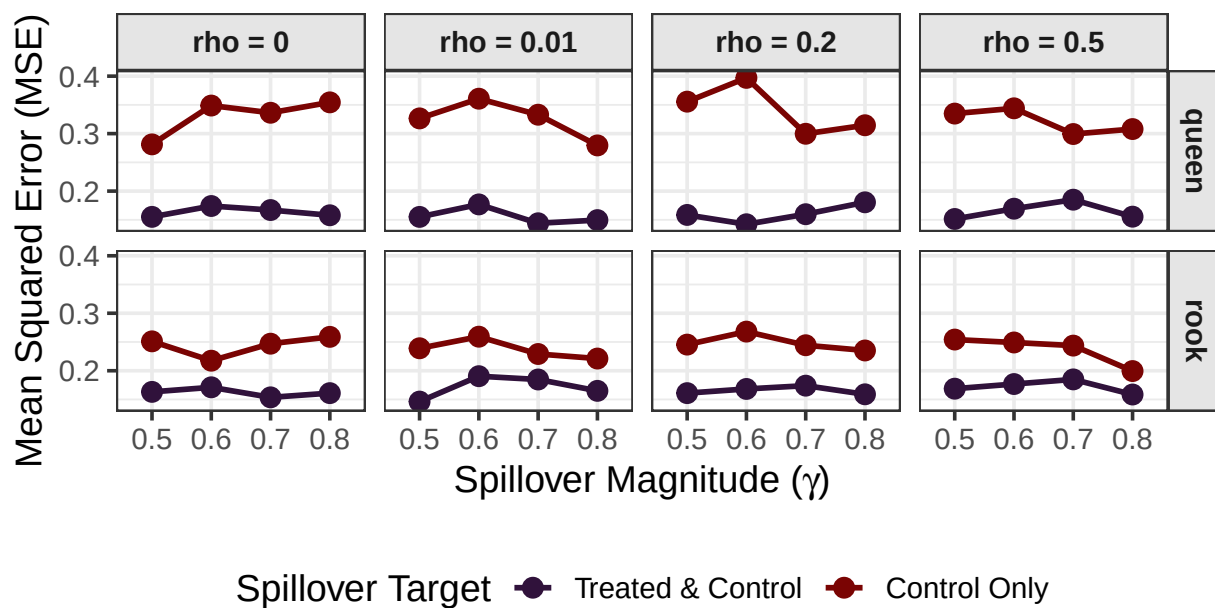


Figure 27: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

\$`Design 3`

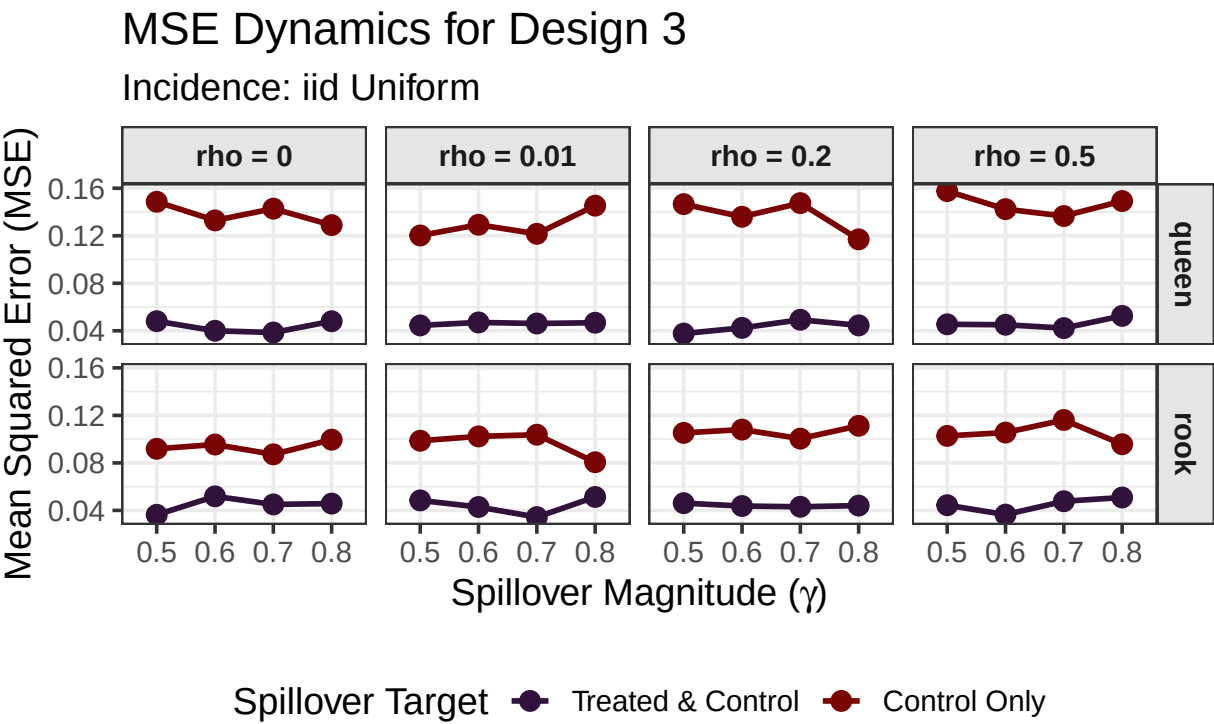


Figure 28: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

\$`Design 4`

\$`Design 5`

\$`Design 6`

\$`Design 7`

\$`Design 8`

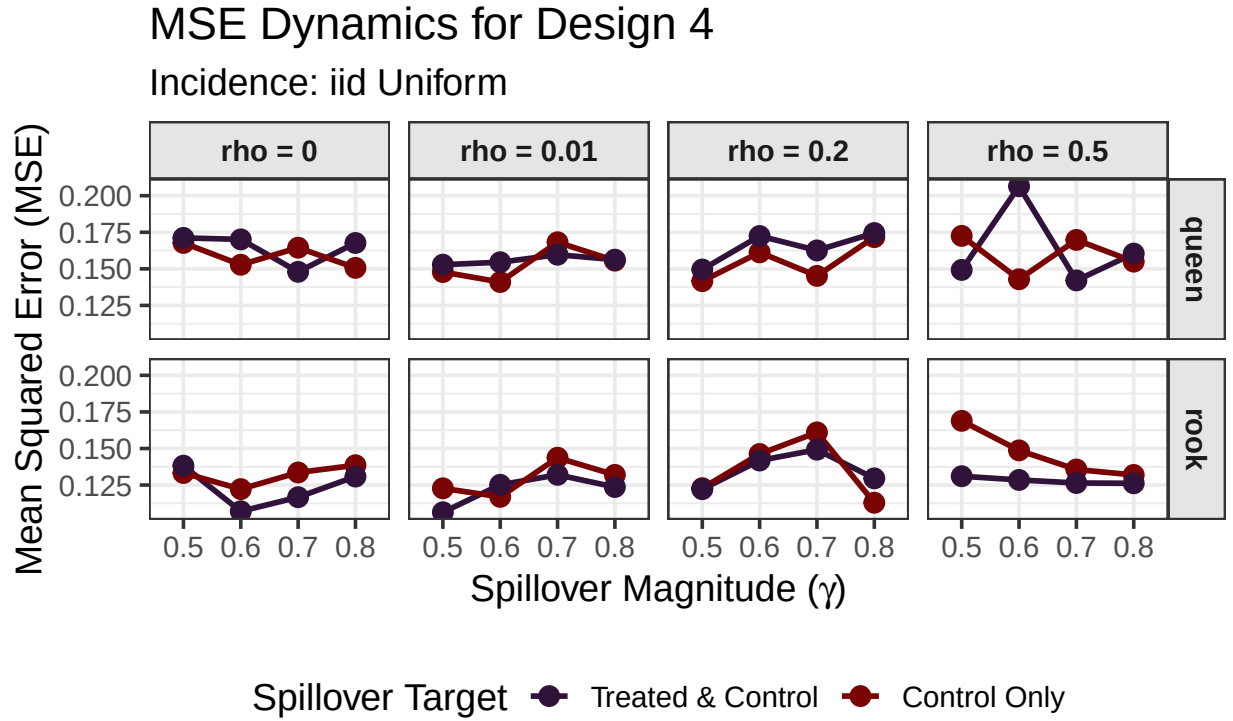


Figure 29: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

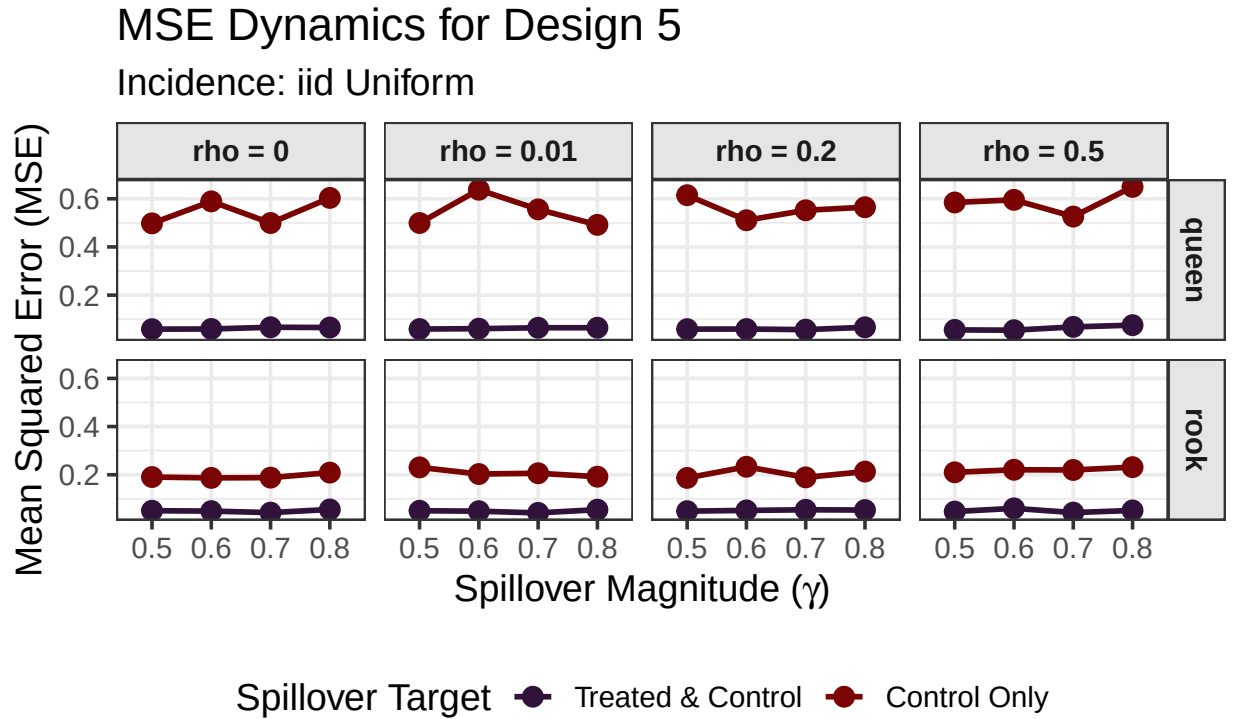


Figure 30: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

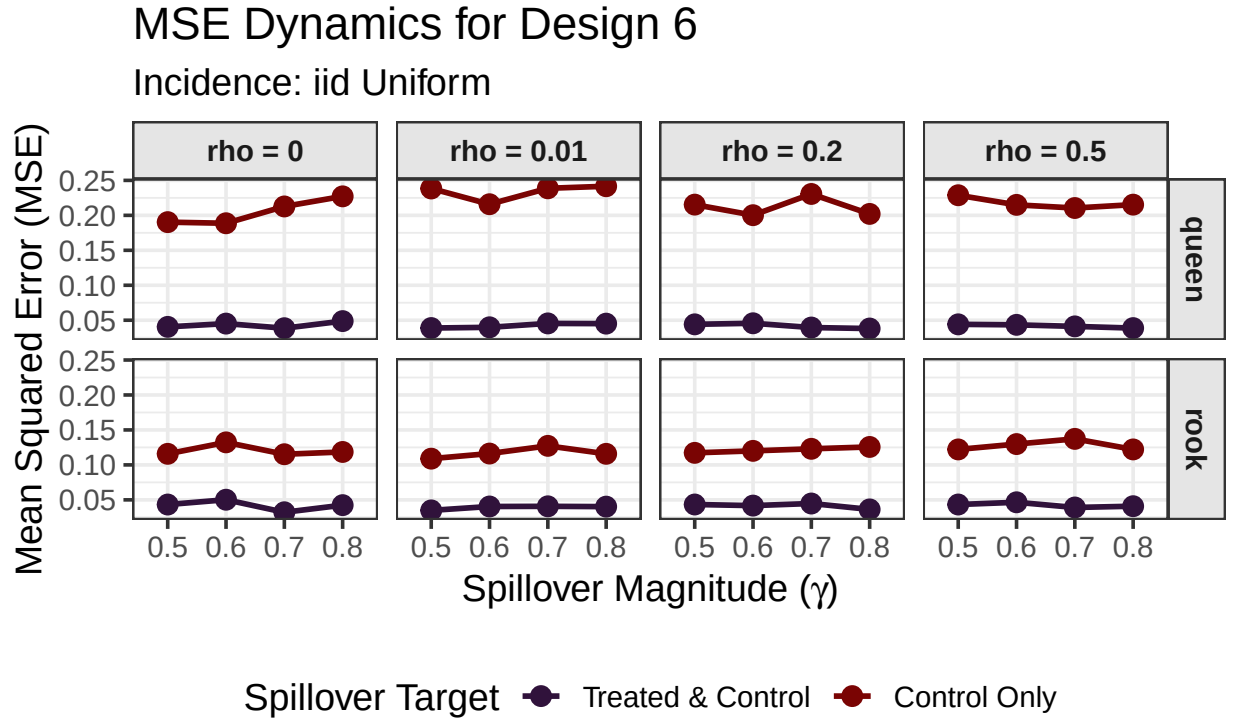


Figure 31: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

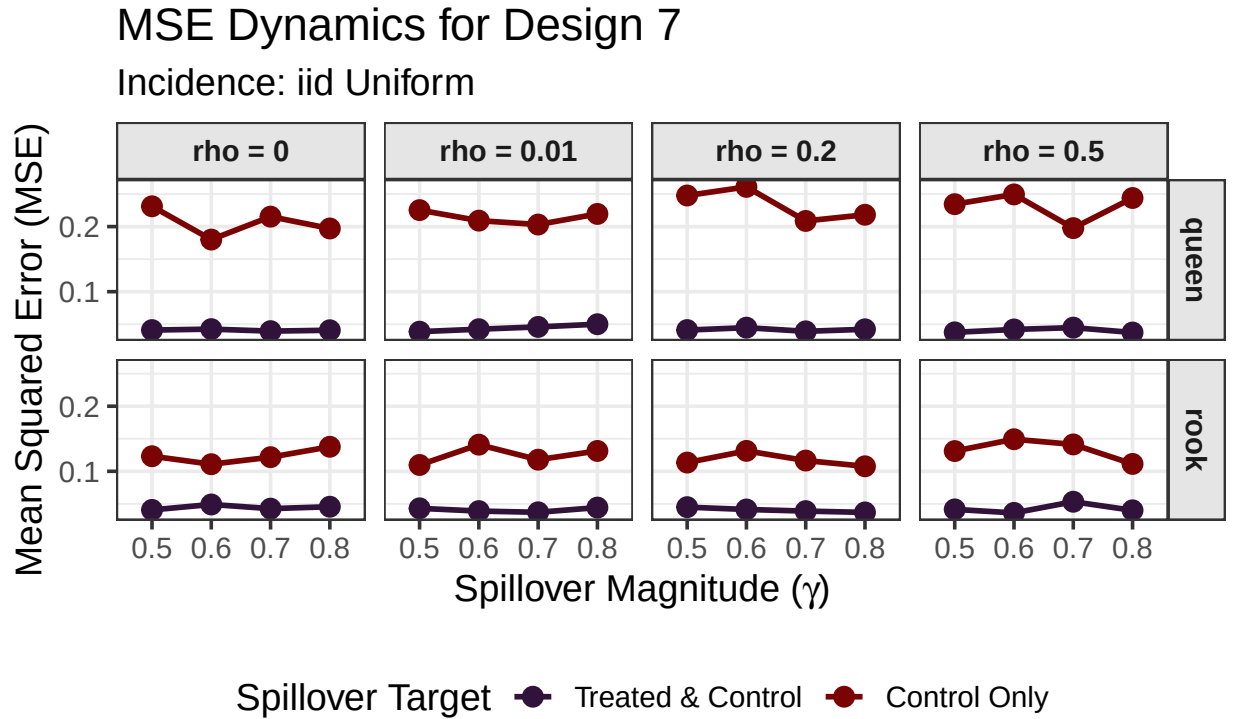


Figure 32: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

MSE Dynamics for Design 8

Incidence: iid Uniform

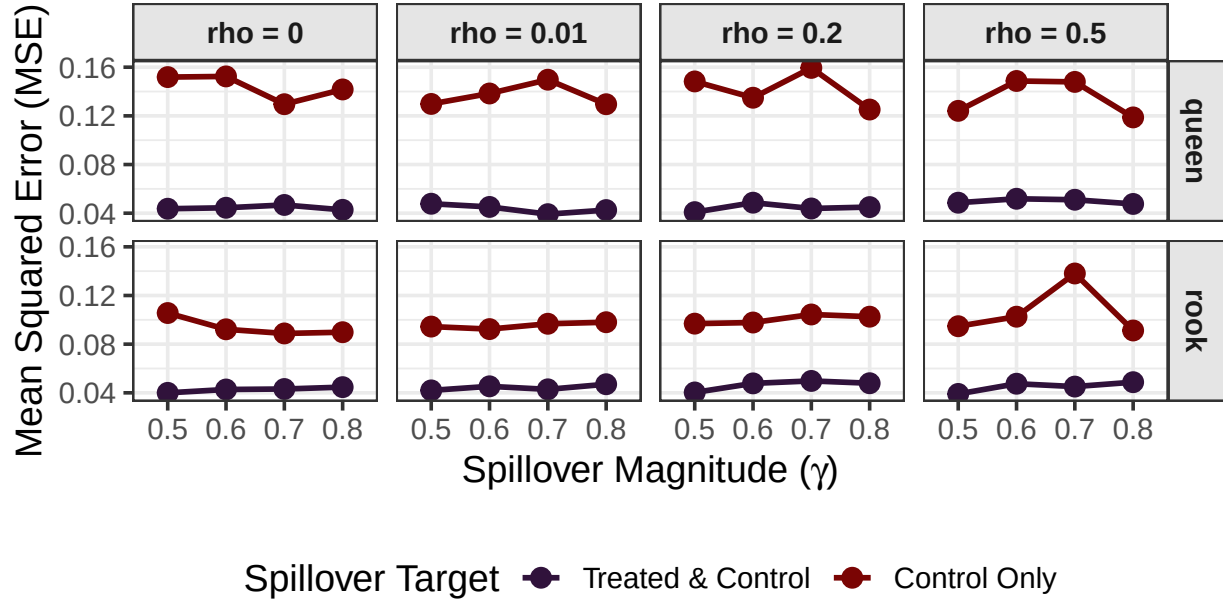


Figure 33: Per-design MSE line plots across gamma levels, for iid Uniform incidence ($\tau = 1.0$).

12.5 Marginal Ranking Tables

12.5.1 iid Uniform

Table 15: iid Uniform | Marginal rankings by Rho (outcome spatial autocorrelation) ($\tau = 1.0$, rook and queen pooled).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D3: Sat. Quadrants	0.0801 (#1)	0.0790 (#1)	0.0827 (#1)	0.0857 (#2)
D8: Inc.-Guided Sat. Q.	0.0813 (#2)	0.0801 (#2)	0.0833 (#2)	0.0841 (#1)
D6: Balanced Quartiles	0.1026 (#3)	0.1080 (#4)	0.1042 (#3)	0.1074 (#3)
D7: Balanced Halves	0.1037 (#4)	0.1061 (#3)	0.1084 (#4)	0.1119 (#4)
D4: Isolation Buffer	0.1446 (#5)	0.1399 (#5)	0.1478 (#5)	0.1497 (#5)
D5: 2x2 Blocking	0.2133 (#6)	0.2163 (#6)	0.2197 (#6)	0.2309 (#7)
D2: High Inc. Focus	0.2249 (#7)	0.2225 (#7)	0.2290 (#7)	0.2239 (#6)
D1: Checkerboard	0.8130 (#8)	0.7638 (#8)	0.8076 (#8)	0.7594 (#8)

Table 16: iid Uniform | Marginal rankings by Gamma (spillover magnitude) ($\tau = 1.0$, rook and queen pooled).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D3: Sat. Quadrants	0.0827 (#2)	0.0813 (#1)	0.0814 (#1)	0.0820 (#2)
D8: Inc.-Guided Sat. Q.	0.0805 (#1)	0.0833 (#2)	0.0860 (#2)	0.0789 (#1)
D6: Balanced Quartiles	0.1043 (#3)	0.1044 (#3)	0.1073 (#4)	0.1062 (#3)
D7: Balanced Halves	0.1090 (#4)	0.1106 (#4)	0.1039 (#3)	0.1065 (#4)

D4: Isolation Buffer	0.1436 (#5)	0.1461 (#5)	0.1474 (#5)	0.1448 (#5)
D5: 2x2 Blocking	0.2153 (#6)	0.2263 (#6)	0.2109 (#6)	0.2277 (#7)
D2: High Inc. Focus	0.2217 (#7)	0.2384 (#7)	0.2241 (#7)	0.2161 (#6)
D1: Checkerboard	0.6944 (#8)	0.7294 (#8)	0.7758 (#8)	0.9442 (#8)

Table 17: iid Uniform | Marginal rankings by Spillover Type ($\tau = 1.0$, rook and queen pooled).

Design	Spillover_Type=both	Spillover_Type=control_only
D3: Sat. Quadrants	0.0447 (#3)	0.1190 (#1)
D8: Inc.-Guided Sat. Q.	0.0451 (#4)	0.1193 (#2)
D6: Balanced Quartiles	0.0418 (#1)	0.1693 (#4)
D7: Balanced Halves	0.0420 (#2)	0.1730 (#5)
D4: Isolation Buffer	0.1448 (#6)	0.1462 (#3)
D5: 2x2 Blocking	0.0562 (#5)	0.3838 (#7)
D2: High Inc. Focus	0.1647 (#7)	0.2855 (#6)
D1: Checkerboard	0.2696 (#8)	1.3023 (#8)

12.5.2 Poisson ($\rho_X = 0.20$)

Table 18: Poisson ($\rho_X = 0.20$) | Marginal rankings by Rho (outcome spatial autocorrelation) ($\tau = 1.0$, rook and queen pooled).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D3: Sat. Quadrants	0.0755 (#1)	0.0760 (#1)	0.0764 (#1)	0.0824 (#1)
D8: Inc.-Guided Sat. Q.	0.0780 (#2)	0.0761 (#2)	0.0860 (#2)	0.0856 (#2)
D6: Balanced Quartiles	0.1041 (#3)	0.1030 (#3)	0.1098 (#4)	0.1090 (#3)
D7: Balanced Halves	0.1125 (#4)	0.1035 (#4)	0.1028 (#3)	0.1122 (#4)
D4: Isolation Buffer	0.1477 (#5)	0.1506 (#5)	0.1459 (#5)	0.1522 (#5)
D2: High Inc. Focus	0.2155 (#6)	0.2105 (#6)	0.2162 (#6)	0.2266 (#6)
D5: 2x2 Blocking	0.2284 (#7)	0.2266 (#7)	0.2248 (#7)	0.2323 (#7)
D1: Checkerboard	0.7953 (#8)	0.7938 (#8)	0.7858 (#8)	0.8103 (#8)

Table 19: Poisson ($\rho_X = 0.20$) | Marginal rankings by Gamma (spillover magnitude) ($\tau = 1.0$, rook and queen pooled).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D3: Sat. Quadrants	0.0777 (#1)	0.0757 (#1)	0.0801 (#1)	0.0767 (#1)
D8: Inc.-Guided Sat. Q.	0.0802 (#2)	0.0828 (#2)	0.0814 (#2)	0.0812 (#2)
D6: Balanced Quartiles	0.1078 (#3)	0.1034 (#3)	0.1059 (#3)	0.1088 (#3)
D7: Balanced Halves	0.1083 (#4)	0.1064 (#4)	0.1070 (#4)	0.1093 (#4)
D4: Isolation Buffer	0.1498 (#5)	0.1484 (#5)	0.1467 (#5)	0.1515 (#5)
D2: High Inc. Focus	0.2184 (#6)	0.2166 (#6)	0.2232 (#6)	0.2106 (#6)
D5: 2x2 Blocking	0.2246 (#7)	0.2250 (#7)	0.2377 (#7)	0.2247 (#7)
D1: Checkerboard	0.7417 (#8)	0.7508 (#8)	0.8159 (#8)	0.8768 (#8)

Table 20: Poisson ($\rho_X = 0.20$) | Marginal rankings by Spillover Type ($\tau = 1.0$, rook and queen pooled).

Design	Spillover_Type=both	Spillover_Type=control_only
--------	---------------------	-----------------------------

D3: Sat. Quadrants	0.0433 (#3)	0.1119 (#1)
D8: Inc.-Guided Sat. Q.	0.0459 (#4)	0.1169 (#2)
D6: Balanced Quartiles	0.0421 (#1)	0.1709 (#4)
D7: Balanced Halves	0.0429 (#2)	0.1726 (#5)
D4: Isolation Buffer	0.1484 (#6)	0.1497 (#3)
D2: High Inc. Focus	0.1602 (#7)	0.2742 (#6)
D5: 2x2 Blocking	0.0555 (#5)	0.4005 (#7)
D1: Checkerboard	0.2694 (#8)	1.3232 (#8)

12.5.3 Poisson (rho_X = 0.50)

Table 21: Poisson (rho X = 0.50) | Marginal rankings by Rho (outcome spatial autocorrelation) (tau = 1.0, rook and queen pooled).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D3: Sat. Quadrants	0.0746 (#1)	0.0799 (#2)	0.0822 (#2)	0.0848 (#2)
D8: Inc.-Guided Sat. Q.	0.0786 (#2)	0.0772 (#1)	0.0820 (#1)	0.0845 (#1)
D7: Balanced Halves	0.1040 (#3)	0.1053 (#3)	0.1074 (#3)	0.1097 (#4)
D6: Balanced Quartiles	0.1101 (#4)	0.1061 (#4)	0.1160 (#4)	0.1082 (#3)
D4: Isolation Buffer	0.1469 (#5)	0.1503 (#5)	0.1462 (#5)	0.1466 (#5)
D2: High Inc. Focus	0.2202 (#6)	0.2247 (#6)	0.2193 (#6)	0.2334 (#7)
D5: 2x2 Blocking	0.2255 (#7)	0.2271 (#7)	0.2211 (#7)	0.2240 (#6)
D1: Checkerboard	0.7828 (#8)	0.7630 (#8)	0.7681 (#8)	0.8052 (#8)

Table 22: Poisson (rho X = 0.50) | Marginal rankings by Gamma (spillover magnitude) (tau = 1.0, rook and queen pooled).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D3: Sat. Quadrants	0.0780 (#1)	0.0773 (#1)	0.0859 (#2)	0.0803 (#1)
D8: Inc.-Guided Sat. Q.	0.0800 (#2)	0.0782 (#2)	0.0819 (#1)	0.0823 (#2)
D7: Balanced Halves	0.1108 (#3)	0.1072 (#3)	0.1049 (#3)	0.1035 (#3)
D6: Balanced Quartiles	0.1118 (#4)	0.1093 (#4)	0.1085 (#4)	0.1108 (#4)
D4: Isolation Buffer	0.1435 (#5)	0.1506 (#5)	0.1508 (#5)	0.1452 (#5)
D2: High Inc. Focus	0.2208 (#6)	0.2214 (#6)	0.2297 (#6)	0.2256 (#7)
D5: 2x2 Blocking	0.2210 (#7)	0.2263 (#7)	0.2300 (#7)	0.2204 (#6)
D1: Checkerboard	0.6930 (#8)	0.7382 (#8)	0.8349 (#8)	0.8529 (#8)

Table 23: Poisson (rho X = 0.50) | Marginal rankings by Spillover Type (tau = 1.0, rook and queen pooled).

Design	Spillover_Type=both	Spillover_Type=control_only
D3: Sat. Quadrants	0.0452 (#3)	0.1155 (#2)
D8: Inc.-Guided Sat. Q.	0.0463 (#4)	0.1149 (#1)
D7: Balanced Halves	0.0417 (#1)	0.1715 (#4)
D6: Balanced Quartiles	0.0430 (#2)	0.1772 (#5)
D4: Isolation Buffer	0.1508 (#6)	0.1442 (#3)
D2: High Inc. Focus	0.1664 (#7)	0.2824 (#6)
D5: 2x2 Blocking	0.0575 (#5)	0.3913 (#7)
D1: Checkerboard	0.2712 (#8)	1.2883 (#8)

12.5.4 Spatial (rho_X = 0.20)

Table 24: Spatial (rho X = 0.20) | Marginal rankings by Rho (outcome spatial autocorrelation) (tau = 1.0, rook and queen pooled).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D8: Inc.-Guided Sat. Q.	0.0791 (#1)	0.0773 (#2)	0.0816 (#2)	0.0798 (#1)
D3: Sat. Quadrants	0.0798 (#2)	0.0754 (#1)	0.0800 (#1)	0.0849 (#2)
D6: Balanced Quartiles	0.1083 (#4)	0.1040 (#3)	0.1081 (#3)	0.1081 (#3)
D7: Balanced Halves	0.1037 (#3)	0.1062 (#4)	0.1126 (#4)	0.1115 (#4)
D4: Isolation Buffer	0.1451 (#5)	0.1398 (#5)	0.1412 (#5)	0.1536 (#5)
D5: 2x2 Blocking	0.2153 (#6)	0.2285 (#6)	0.2150 (#6)	0.2217 (#6)
D2: High Inc. Focus	0.2415 (#7)	0.2424 (#7)	0.2379 (#7)	0.2376 (#7)
D1: Checkerboard	0.7721 (#8)	0.8015 (#8)	0.7681 (#8)	0.7578 (#8)

Table 25: Spatial (rho X = 0.20) | Marginal rankings by Gamma (spillover magnitude) (tau = 1.0, rook and queen pooled).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D8: Inc.-Guided Sat. Q.	0.0792 (#2)	0.0781 (#1)	0.0834 (#2)	0.0772 (#1)
D3: Sat. Quadrants	0.0782 (#1)	0.0820 (#2)	0.0798 (#1)	0.0801 (#2)
D6: Balanced Quartiles	0.1044 (#3)	0.1081 (#3)	0.1091 (#3)	0.1070 (#3)
D7: Balanced Halves	0.1073 (#4)	0.1089 (#4)	0.1093 (#4)	0.1086 (#4)
D4: Isolation Buffer	0.1489 (#5)	0.1494 (#5)	0.1430 (#5)	0.1385 (#5)
D5: 2x2 Blocking	0.2207 (#6)	0.2087 (#6)	0.2221 (#6)	0.2290 (#6)
D2: High Inc. Focus	0.2409 (#7)	0.2463 (#7)	0.2309 (#7)	0.2412 (#7)
D1: Checkerboard	0.6726 (#8)	0.7076 (#8)	0.8169 (#8)	0.9024 (#8)

Table 26: Spatial (rho X = 0.20) | Marginal rankings by Spillover Type (tau = 1.0, rook and queen pooled).

Design	Spillover_Type=both	Spillover_Type=control_only
D8: Inc.-Guided Sat. Q.	0.0451 (#4)	0.1138 (#1)
D3: Sat. Quadrants	0.0445 (#3)	0.1155 (#2)
D6: Balanced Quartiles	0.0421 (#1)	0.1722 (#4)
D7: Balanced Halves	0.0430 (#2)	0.1740 (#5)
D4: Isolation Buffer	0.1429 (#6)	0.1470 (#3)
D5: 2x2 Blocking	0.0571 (#5)	0.3832 (#7)
D2: High Inc. Focus	0.1673 (#7)	0.3124 (#6)
D1: Checkerboard	0.2712 (#8)	1.2786 (#8)

12.5.5 Spatial (rho_X = 0.50)

Table 27: Spatial (rho X = 0.50) | Marginal rankings by Rho (outcome spatial autocorrelation) (tau = 1.0, rook and queen pooled).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D8: Inc.-Guided Sat. Q.	0.0775 (#1)	0.0789 (#1)	0.0821 (#2)	0.0900 (#2)
D3: Sat. Quadrants	0.0804 (#2)	0.0816 (#2)	0.0803 (#1)	0.0865 (#1)
D7: Balanced Halves	0.1035 (#4)	0.1013 (#3)	0.1052 (#3)	0.1098 (#3)

D6: Balanced Quartiles	0.1027 (#3)	0.1069 (#4)	0.1053 (#4)	0.1134 (#4)
D4: Isolation Buffer	0.1435 (#5)	0.1433 (#5)	0.1450 (#5)	0.1507 (#5)
D2: High Inc. Focus	0.2173 (#6)	0.2215 (#6)	0.2262 (#7)	0.2258 (#7)
D5: 2x2 Blocking	0.2227 (#7)	0.2302 (#7)	0.2251 (#6)	0.2253 (#6)
D1: Checkerboard	0.7757 (#8)	0.7660 (#8)	0.7795 (#8)	0.7994 (#8)

Table 28: Spatial ($\rho_X = 0.50$) | Marginal rankings by Gamma (spillover magnitude) ($\tau = 1.0$, rook and queen pooled).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D8: Inc.-Guided Sat. Q.	0.0827 (#1)	0.0805 (#2)	0.0824 (#1)	0.0829 (#2)
D3: Sat. Quadrants	0.0854 (#2)	0.0783 (#1)	0.0829 (#2)	0.0822 (#1)
D7: Balanced Halves	0.1063 (#3)	0.1058 (#3)	0.1049 (#3)	0.1028 (#4)
D6: Balanced Quartiles	0.1106 (#4)	0.1082 (#4)	0.1072 (#4)	0.1023 (#3)
D4: Isolation Buffer	0.1408 (#5)	0.1447 (#5)	0.1472 (#5)	0.1499 (#5)
D2: High Inc. Focus	0.2265 (#7)	0.2222 (#7)	0.2245 (#6)	0.2176 (#6)
D5: 2x2 Blocking	0.2203 (#6)	0.2218 (#6)	0.2364 (#7)	0.2249 (#7)
D1: Checkerboard	0.7065 (#8)	0.7120 (#8)	0.8151 (#8)	0.8869 (#8)

Table 29: Spatial ($\rho_X = 0.50$) | Marginal rankings by Spillover Type ($\tau = 1.0$, rook and queen pooled).

Design	Spillover_Type=both	Spillover_Type=control_only
D8: Inc.-Guided Sat. Q.	0.0451 (#4)	0.1192 (#1)
D3: Sat. Quadrants	0.0440 (#3)	0.1204 (#2)
D7: Balanced Halves	0.0418 (#2)	0.1681 (#4)
D6: Balanced Quartiles	0.0415 (#1)	0.1726 (#5)
D4: Isolation Buffer	0.1437 (#6)	0.1476 (#3)
D2: High Inc. Focus	0.1550 (#7)	0.2904 (#6)
D5: 2x2 Blocking	0.0558 (#5)	0.3959 (#7)
D1: Checkerboard	0.2659 (#8)	1.2943 (#8)

13 Results: Best Design by Parameter Combination (Q3)

13.1 Rho x Gamma Best-Design Maps

13.2 Rho x Spillover Type Best-Design Maps

14 Per-Config Narrative Findings

The findings below are generated from the $\tau = 1.0$ results with rook and queen pooled. In every configuration the top two designs are Saturation Quadrants and Incidence-Guided Saturation Quadrants, which are statistically indistinguishable (Section 10); which of the two is listed as

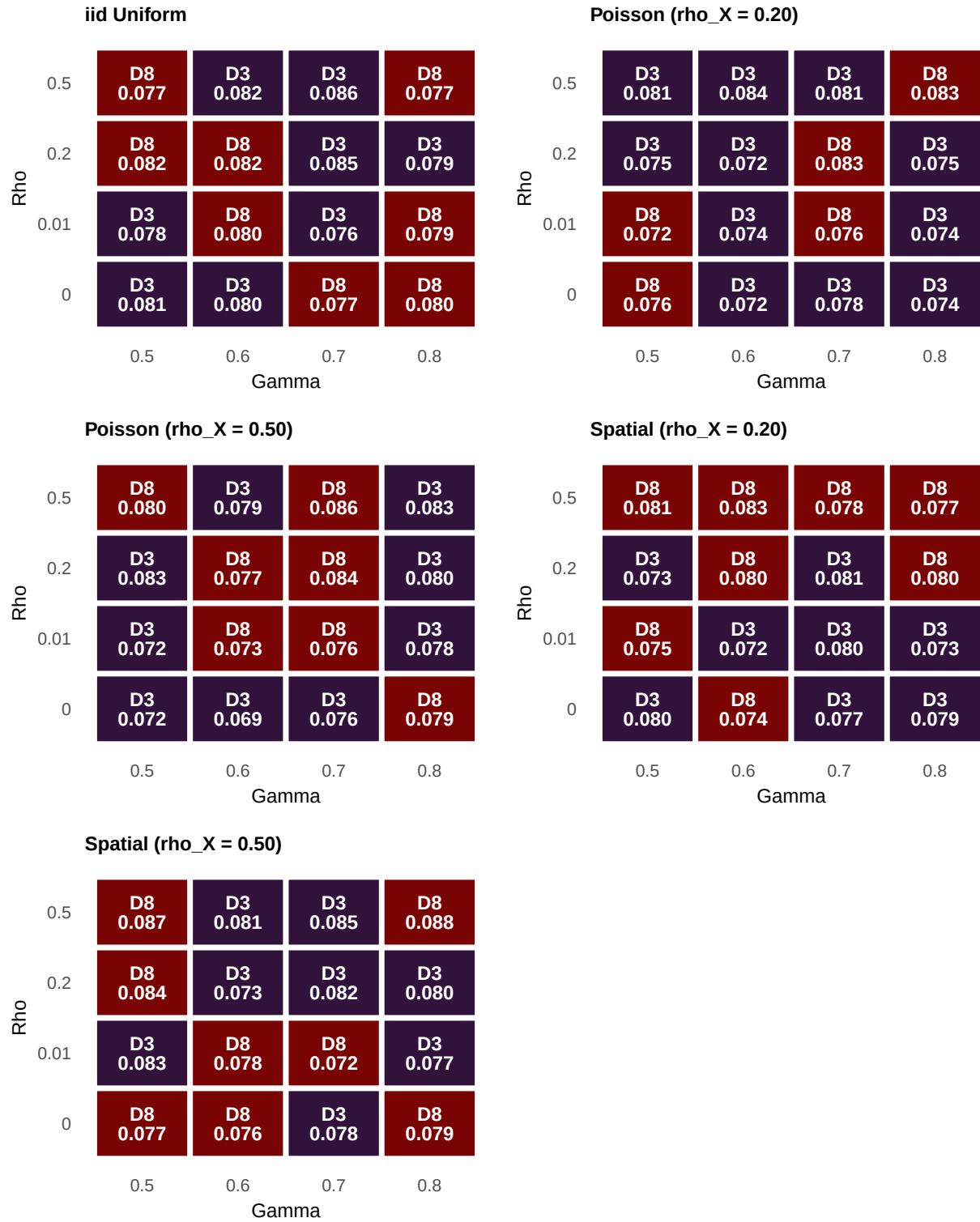


Figure 34: Best design heatmap: Rho vs. Gamma for each incidence config ($\tau = 1.0$, rook and queen pooled). Each tile shows which design achieves the lowest average MSE.

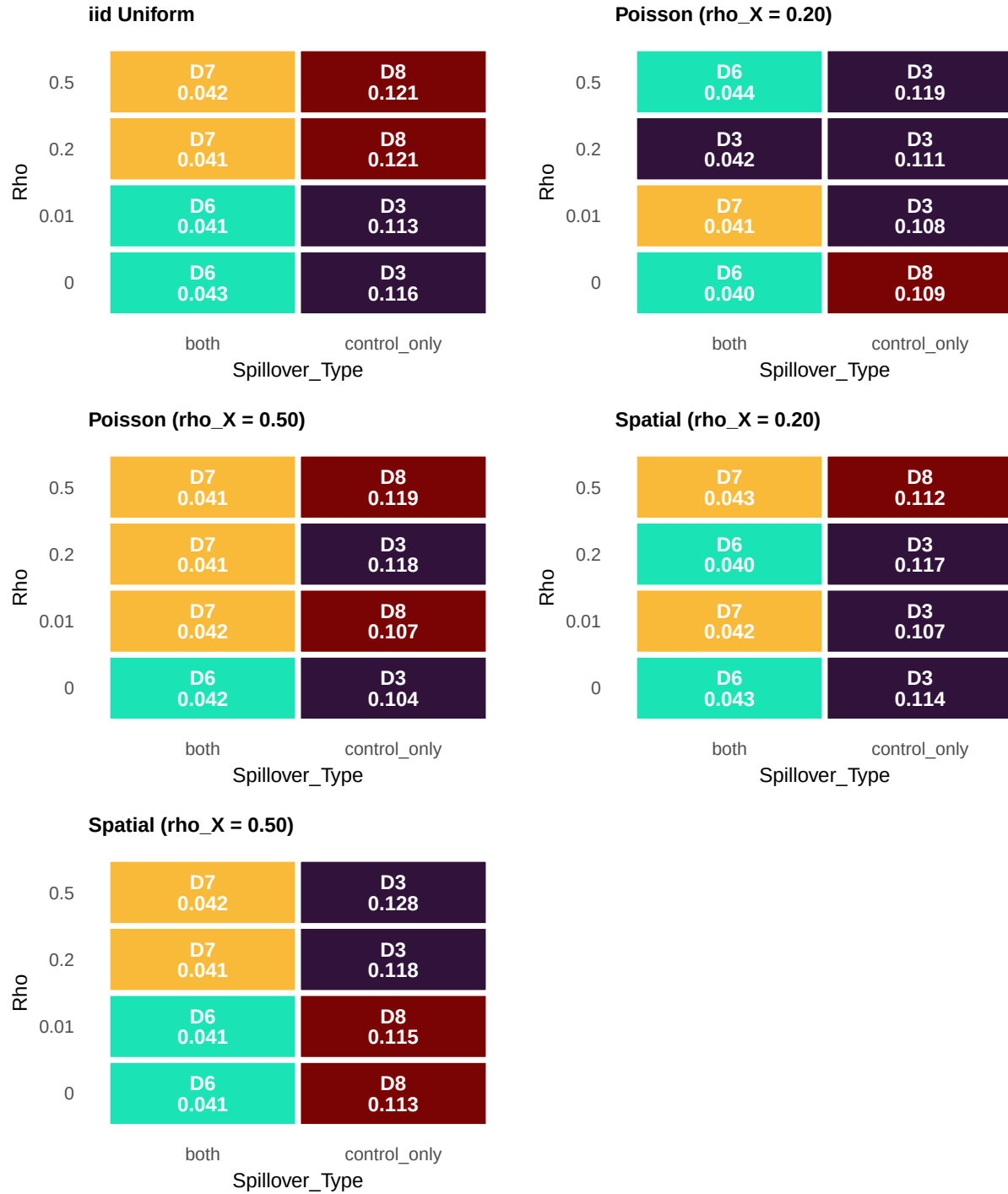


Figure 35: Best design heatmap: Rho vs. Spillover Type for each incidence config ($\tau = 1.0$, rook and queen pooled). Reveals the key spillover-type interaction.

the “winner” depends on differences in average MSE of at most 0.004, smaller than the typical per-scenario Monte Carlo SE of their MSE (about 0.007; Section 18). The Checkerboard coverage quoted here is pooled; it is near nominal under queen and about 0.15 under rook.

14.1 iid Uniform

Overall Winner: D3: Sat. Quadrants achieves the lowest average MSE (0.0818). Runner-up: D8: Inc.-Guided Sat. Q. (0.0822). The winner dominates in 50.0% of individual scenarios.

Robustness: D3: Sat. Quadrants shows some rank variation across parameter levels.

Coverage: D3: Sat. Quadrants achieves coverage of 0.935; worst design (D1: Checkerboard) achieves only 0.545.

Recommendation: Under iid Uniform incidence, select D3: Sat. Quadrants. It provides 9.6x lower MSE than D1: Checkerboard and near-nominal coverage.

14.2 Poisson ($\rho_X = 0.20$)

Overall Winner: D3: Sat. Quadrants achieves the lowest average MSE (0.0776). Runner-up: D8: Inc.-Guided Sat. Q. (0.0814). The winner dominates in 59.4% of individual scenarios.

Robustness: D3: Sat. Quadrants holds rank 1 at every level of Rho and Gamma.

Coverage: D3: Sat. Quadrants achieves coverage of 0.942; worst design (D1: Checkerboard) achieves only 0.542.

Recommendation: Under Poisson ($\rho_X = 0.20$) incidence, select D3: Sat. Quadrants. It provides 10.3x lower MSE than D1: Checkerboard and near-nominal coverage.

14.3 Poisson ($\rho_X = 0.50$)

Overall Winner: D3: Sat. Quadrants achieves the lowest average MSE (0.0804). Runner-up: D8: Inc.-Guided Sat. Q. (0.0806). The winner dominates in 45.3% of individual scenarios.

Robustness: D3: Sat. Quadrants shows some rank variation across parameter levels.

Coverage: D3: Sat. Quadrants achieves coverage of 0.939; worst design (D1: Checkerboard) achieves only 0.544.

Recommendation: Under Poisson ($\rho_X = 0.50$) incidence, select D3: Sat. Quadrants. It provides 9.7x lower MSE than D1: Checkerboard and near-nominal coverage.

14.4 Spatial ($\rho_X = 0.20$)

Overall Winner: D8: Inc.-Guided Sat. Q. achieves the lowest average MSE (0.0795). Runner-up: D3: Sat. Quadrants (0.0800). The winner dominates in 48.4% of individual scenarios.

Robustness: D8: Inc.-Guided Sat. Q. shows some rank variation across parameter levels.

Coverage: D8: Inc.-Guided Sat. Q. achieves coverage of 0.942; worst design (D1: Checkerboard) achieves only 0.547.

Recommendation: Under Spatial ($\rho_X = 0.20$) incidence, select D8: Inc.-Guided Sat. Q.. It provides 9.8x lower MSE than D1: Checkerboard and near-nominal coverage.

14.5 Spatial ($\rho_X = 0.50$)

Overall Winner: D8: Inc.-Guided Sat. Q. achieves the lowest average MSE (0.0821). Runner-up: D3: Sat. Quadrants (0.0822). The winner dominates in 46.9% of individual scenarios.

Robustness: D8: Inc.-Guided Sat. Q. shows some rank variation across parameter levels.

Coverage: D8: Inc.-Guided Sat. Q. achieves coverage of 0.937; worst design (D1: Checkerboard) achieves only 0.548.

Recommendation: Under Spatial ($\rho_X = 0.50$) incidence, select D8: Inc.-Guided Sat. Q.. It provides 9.5x lower MSE than D1: Checkerboard and near-nominal coverage.

15 Practical Design Selection Guide

Table 30: Practical design selection guide ($\tau = 1.0$, rook and queen pooled).

Situation	Recommended Design	Justification
Default (uncertain parameters)	D3: Sat. Quadrants	Ranks 1st in 3 of 5 configs across all Gamma levels
Spillover is 'both' AND $\rho \leq 0.01$	D6: Balanced Quartiles	Winner shifts at this corner; verify from $\rho \times \text{Spillover_Type}$ heatmap
ANY scenario	Avoid D1: Checkerboard	Pooled coverage 0.54–0.55 (rook: τ not identified, coverage 0.15; queen: near nominal but largest MSE); pooled MSE 9.5–10.3x higher than winner
ANY scenario	Avoid D5: 2x2 Blocking under queen contiguity	Queen contiguity amplifies 2x2 block effects

16 Subgroup Robustness Profiles

16.1 iid Uniform

Table 31: MLE robustness profile: iid Uniform ($\tau = 1.0$, rook and queen pooled)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0344	0.0451	0.0665	0.1163	0.1576	0.0713
Design 8	0.0391	0.0451	0.0703	0.1201	0.1594	0.0750
Design 6	0.0322	0.0416	0.0795	0.1501	0.2416	0.1085
Design 7	0.0361	0.0415	0.0805	0.1570	0.2610	0.1155
Design 5	0.0416	0.0558	0.1311	0.2978	0.6493	0.2420
Design 4	0.1064	0.1310	0.1457	0.1606	0.2065	0.0296
Design 2	0.1423	0.1626	0.1949	0.2709	0.3970	0.1083
Design 1	0.0547	0.2181	0.4698	0.9830	2.8430	0.7649

16.2 Poisson ($\rho_X = 0.20$)

Table 32: MLE robustness profile: Poisson ($\rho_X = 0.20$) ($\tau = 1.0$, rook and queen pooled)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0330	0.0438	0.0642	0.1063	0.1525	0.0625
Design 8	0.0369	0.0455	0.0666	0.1188	0.1612	0.0733
Design 7	0.0330	0.0427	0.0794	0.1598	0.2429	0.1171
Design 6	0.0346	0.0420	0.0813	0.1491	0.2467	0.1071
Design 5	0.0424	0.0545	0.1253	0.3070	0.6581	0.2525
Design 4	0.1159	0.1379	0.1489	0.1613	0.1870	0.0234
Design 2	0.1431	0.1596	0.1935	0.2754	0.3601	0.1158
Design 1	0.0514	0.2365	0.4786	0.9893	2.4848	0.7529

16.3 Poisson ($\rho_X = 0.50$)

Table 33: MLE robustness profile: Poisson ($\rho_X = 0.50$) ($\tau = 1.0$, rook and queen pooled)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0391	0.0446	0.0648	0.1143	0.1569	0.0696
Design 8	0.0342	0.0459	0.0685	0.1097	0.1563	0.0639
Design 6	0.0360	0.0436	0.0776	0.1702	0.2740	0.1266
Design 7	0.0349	0.0419	0.0784	0.1640	0.2482	0.1221
Design 5	0.0445	0.0570	0.1207	0.2959	0.6410	0.2389
Design 4	0.1160	0.1337	0.1461	0.1592	0.1950	0.0254
Design 2	0.1389	0.1664	0.2043	0.2915	0.3549	0.1251
Design 1	0.0539	0.2285	0.4739	0.9710	2.3586	0.7426

16.4 Spatial ($\rho_X = 0.20$)

Table 34: MLE robustness profile: Spatial ($\rho_X = 0.20$) ($\tau = 1.0$, rook and queen pooled)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0370	0.0447	0.0676	0.1118	0.1654	0.0671
Design 8	0.0362	0.0442	0.0720	0.1142	0.1518	0.0700
Design 6	0.0368	0.0417	0.0797	0.1583	0.2512	0.1166
Design 7	0.0341	0.0434	0.0830	0.1588	0.2630	0.1154
Design 5	0.0394	0.0587	0.1272	0.2933	0.6650	0.2347
Design 4	0.1078	0.1318	0.1434	0.1577	0.1985	0.0259
Design 2	0.1377	0.1686	0.2010	0.3009	0.4239	0.1323
Design 1	0.0549	0.2126	0.4762	0.9709	2.2620	0.7583

16.5 Spatial ($\rho_X = 0.50$)

Table 35: MLE robustness profile: Spatial ($\rho_X = 0.50$) ($\tau = 1.0$, rook and queen pooled)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0353	0.0434	0.0707	0.1126	0.1689	0.0692
Design 8	0.0361	0.0454	0.0707	0.1171	0.1688	0.0717
Design 6	0.0340	0.0417	0.0779	0.1568	0.2904	0.1151
Design 7	0.0351	0.0416	0.0780	0.1481	0.2664	0.1065
Design 5	0.0406	0.0550	0.1302	0.2975	0.7190	0.2425
Design 4	0.1057	0.1318	0.1441	0.1622	0.1840	0.0304
Design 2	0.1378	0.1562	0.2006	0.2853	0.3807	0.1291
Design 1	0.0532	0.2236	0.4639	0.9839	2.4540	0.7603

16.6 MSE Heatmap

16.7 Design Rank Frequency

Table 36: MLE: Design rank frequency (%) for iid Uniform incidence ($\tau = 1.0$, rook and queen pooled). Each cell = % of scenarios where design achieved that rank.

Design	Rank_5	Rank_6	Rank_8	Rank_7	Rank_1	Rank_2	Rank_3	Rank_4
Design 8	1.6	0.0	0.0	0.0	32.8	26.6	17.2	21.9
Design 3	3.1	0.0	0.0	0.0	31.2	31.2	15.6	18.8
Design 7	18.8	0.0	0.0	0.0	17.2	18.8	20.3	25.0
Design 6	15.6	0.0	0.0	0.0	15.6	18.8	23.4	26.6
Design 4	17.2	25.0	12.5	12.5	3.1	3.1	20.3	6.2
Design 1	10.9	14.1	75.0	0.0	0.0	0.0	0.0	0.0
Design 2	0.0	26.6	12.5	60.9	0.0	0.0	0.0	0.0

Design	Rank_5	Rank_6	Rank_8	Rank_7	Rank_1	Rank_2	Rank_3	Rank_4
Design 5	32.8	34.4	0.0	26.6	0.0	1.6	3.1	1.6

17 Conclusions and Future Work

The simulation study evaluated 8 treatment assignment designs across 12,800 scenarios; the findings below use the 2,560 $\tau = 1.0$ scenarios with rook and queen pooled unless stated.

Key findings:

1. **MLE achieves near-nominal coverage except for Checkerboard under rook.** Coverage is about 0.94 for every design except Checkerboard $\times$ rook (0.15), where τ is not identified; the pooled average over all scenarios, 0.89, is pulled down by that cell. The remaining shortfall of about one point below 0.95 is a finite-sample property of the Wald interval. Coverage does not separate the designs; MSE does.
2. **Design 3 has the lowest pooled average MSE** (0.08, coverage 0.938), statistically tied with the runner-up; Saturation Quadrants and Incidence-Guided Saturation Quadrants are indistinguishable. Under queen alone their MSEs are 0.090 and 0.091.
3. **Design 1 is consistently worst.** Highest pooled MSE (0.783) and lowest pooled coverage (0.545). Under rook its perfect alternation makes $\mathbf{WZ} = \mathbf{1} - \mathbf{Z}$, so τ is not identified (bias about $-\gamma$); under queen it is almost unbiased but very variable (MSE 1.087), almost entirely from control-only spillover, where the Spill covariate is nearly collinear with Z (MSE 2.11 vs 0.06 under bidirectional spillover).
4. **Spillover type is the key moderator (pooled).** Under control-only spillover, D8: Inc.-Guided Sat. Q. dominates. Under bidirectional spillover, D6: Balanced Quartiles becomes competitive.
5. **Design rankings are stable across incidence modes and Gamma levels.** The ordering is largely consistent, and spillover magnitude does not affect design choice.

17.1 Limitations and Planned Extensions

- **Oracle MLE:** The MLE model uses the true spillover covariate. A non-oracle sensitivity analysis ($Y \sim Z + X$) was run on the same outcomes (`results/sim_data/sim_results_MLEnonoracle_tau_sweep`) but is not reported here.
- **Tau sensitivity:** τ was swept across $\{0.8, 1.0, 1.5, 2.0, 3.0\}$ in the completed tau-sweep run (12,800 scenarios). See Section 18 for power curves and rank-stability results. Primary results use $\tau = 1.0$.
- **Equal populations:** Poisson mode uses $P_i = 100,000$ for all clusters.
- **Grid sensitivity:** Results use $G = 10$ ($N = 100$) only.

Heatmap of Error by Spatial Dependence & Neighbo Incidence: iid Uniform

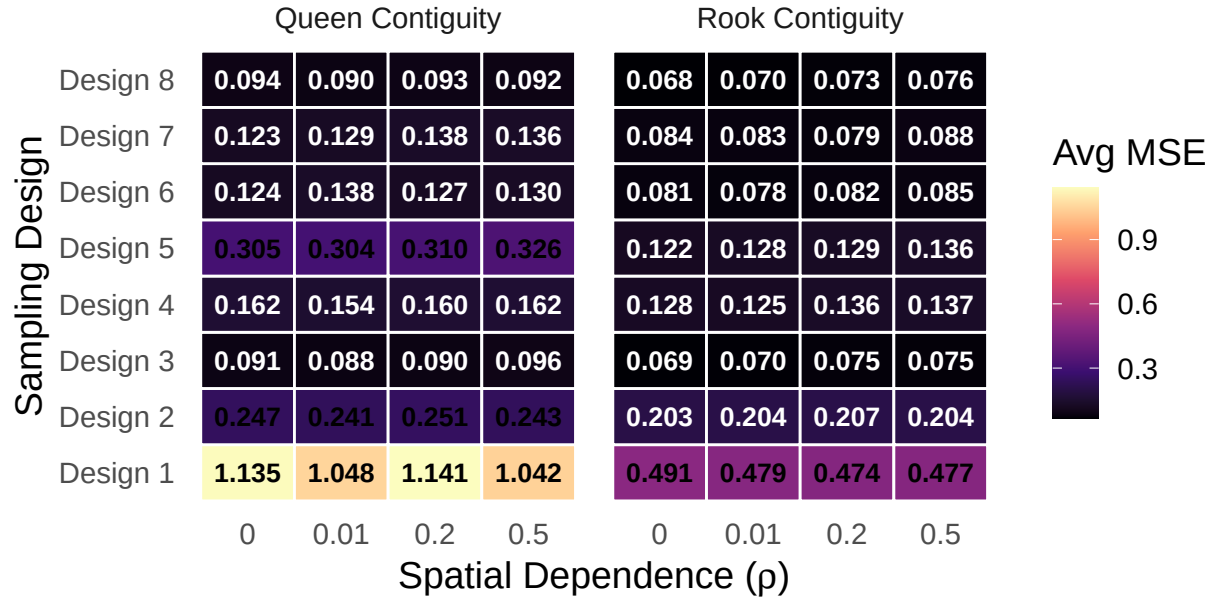


Figure 36: MLE MSE heatmap for iid Uniform incidence ($\tau = 1.0$, rook and queen pooled).

18 Tau Sensitivity Analysis

The completed tau-sweep simulation (12,800 scenarios) swept $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$. This section asks: do the design rankings established at $\tau = 1.0$ hold across other effect sizes, and when do designs achieve adequate statistical power?

18.1 MSE Across Tau Values

18.2 Power Curves

18.3 Coverage Across Tau Values

18.4 Rank Stability Across Tau Values

18.5 Monte Carlo Standard Errors

The `N_Valid_Est` column records the number of valid MLE estimates per scenario. All 2,560 primary-scenario ($\tau = 1.0$) rows have `N_Valid_Est` = 250 (full convergence), and this holds across all 12,800 tau-sweep scenarios (`Fail_Rate` = 0.0 everywhere), so reported MSE and coverage differences are not driven by MLE failures. (Checkerboard $\times$ rook fits are all flagged as aliased: they converge, but τ is not identified.)

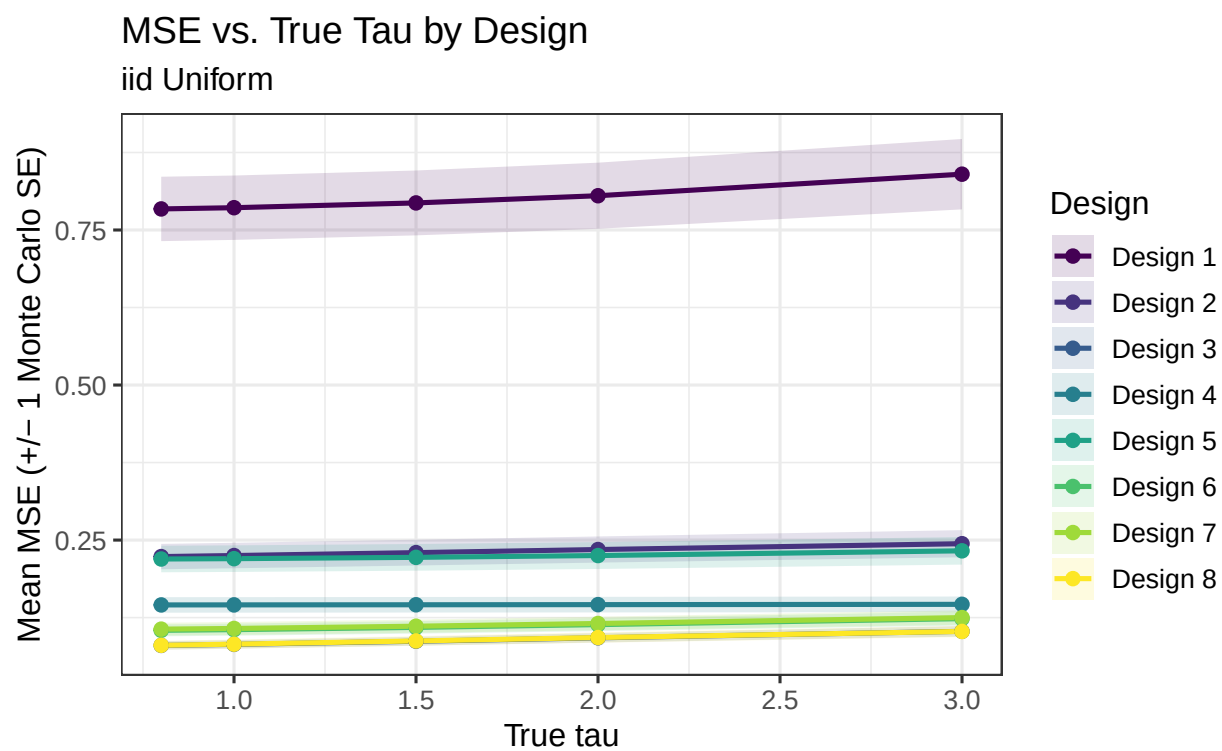


Figure 37: MSE by design across tau values (iid Uniform incidence, rook and queen pooled). Bands show ± 1 surface-level Monte Carlo SE. Checkerboard is worst at every tau.

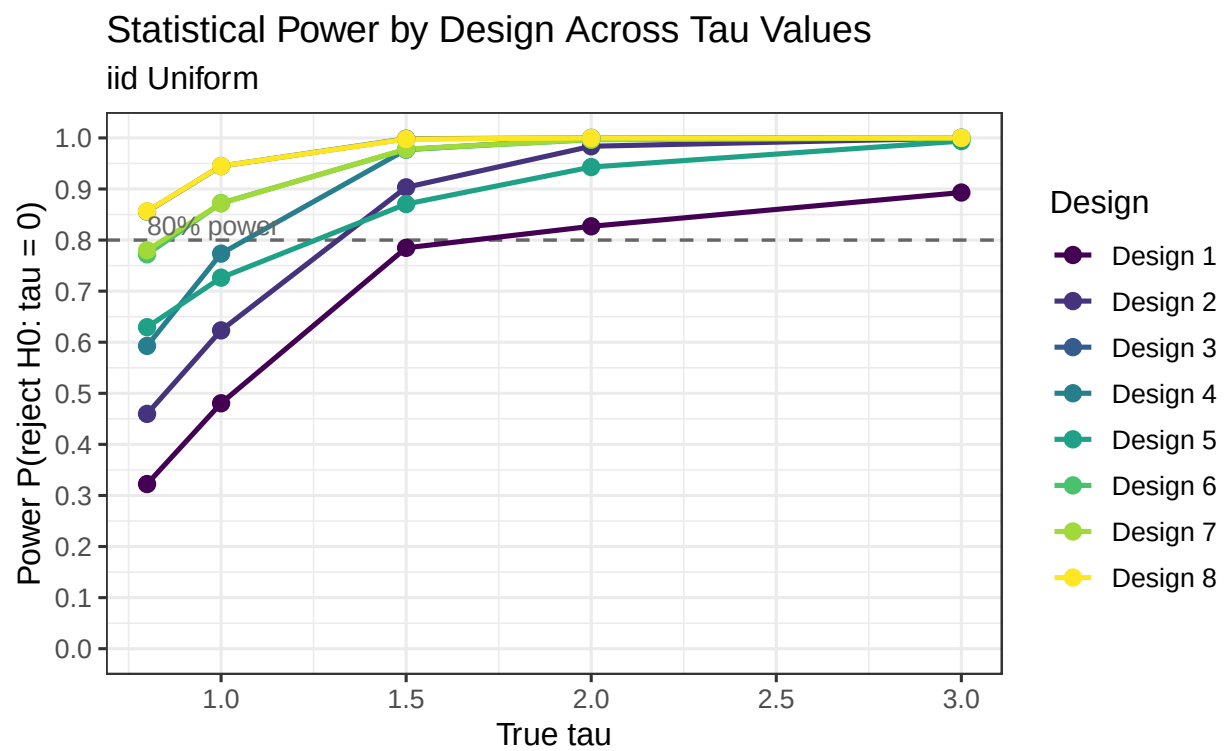


Figure 38: Statistical power $P(\text{reject } H_0: \tau=0)$ by design across tau values (iid Uniform, rook and queen pooled). 80% power reference shown. Efficient designs reach 80% power at smaller effect sizes.

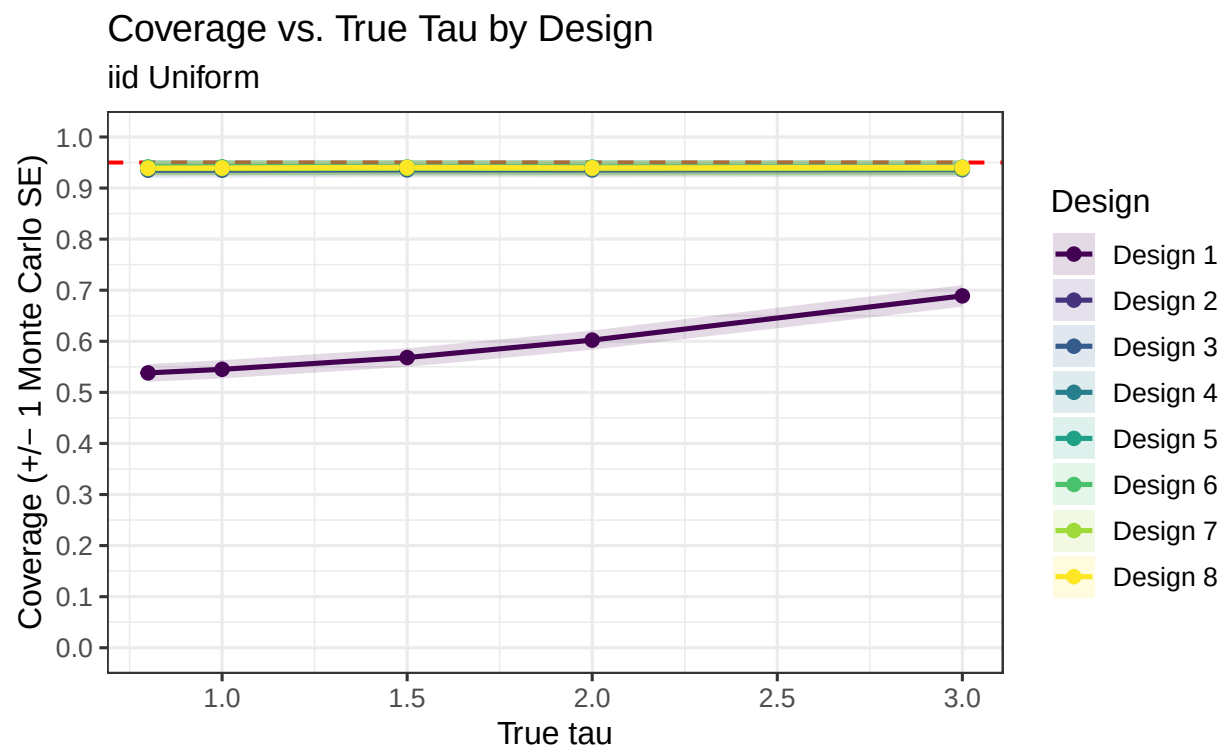


Figure 39: MLE coverage by design across tau values (iid Uniform, rook and queen pooled). Nominal 95% reference shown. Valid designs maintain coverage independent of effect size.

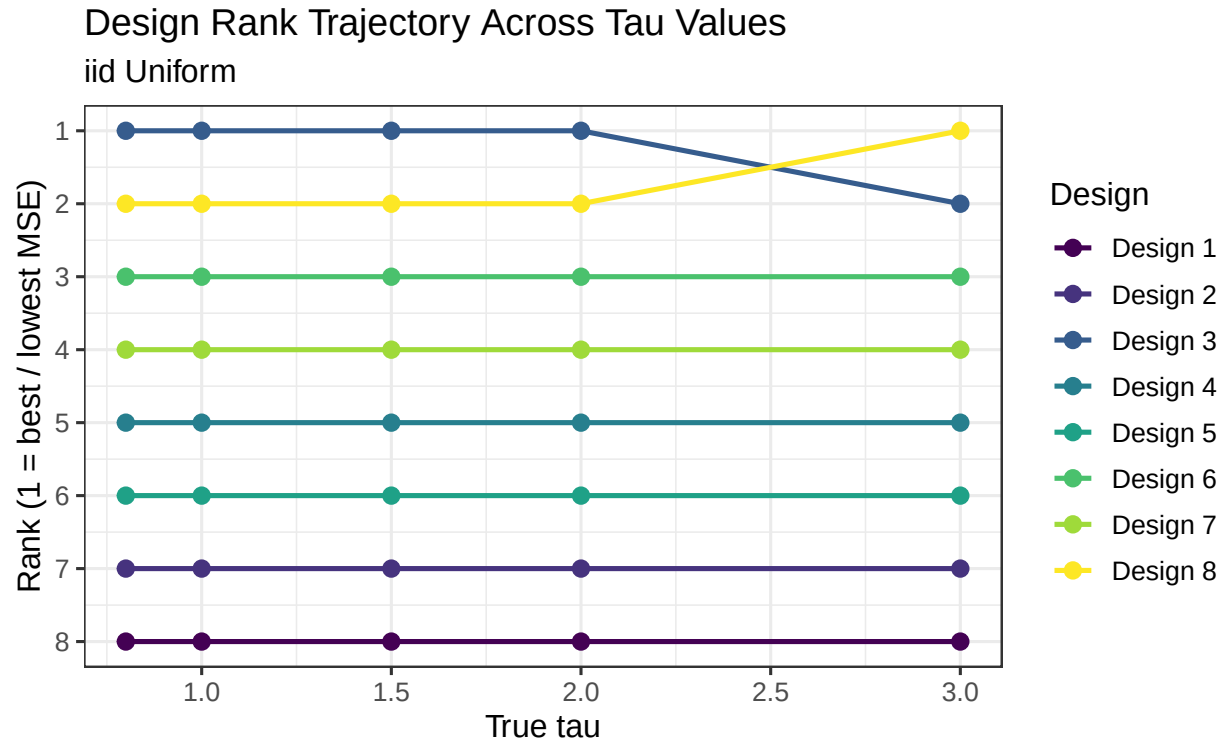


Figure 40: Design rank trajectories across tau values (iid Uniform, rook and queen pooled; rank 1 = best MSE). Stable trajectories show that the top ranking of Saturation Quadrants and Incidence-Guided Saturation Quadrants is not an artifact of the primary effect size. Each tau reuses the same design and noise draws, so the tau levels are not independent confirmations.

The Monte Carlo (MC) standard errors shown below are the stored surface-level SEs (SE_MSE): for each scenario, the MSE is computed separately on each of the $K = 10$ incidence surfaces, and $SE = \text{sd}(MSE_1, \dots, MSE_{10})/\sqrt{10}$. The surface is the primary sampling unit, so the 250 fits are not treated as independent.

Table 37: Monte Carlo standard errors for MSE, $\tau = 1.0$ slice (2,560 scenarios, rook and queen pooled). MC SE is the surface-level SE (sd of the 10 surface-level MSEs / $\sqrt{10}$), averaged over scenarios. The SEs are small relative to the gaps between most designs, but not relative to the gap between Saturation Quadrants and Incidence-Guided Saturation Quadrants, or between Balanced Quartiles and Balanced Halves. These are per-design SEs, not SEs of paired differences.

Design	Mean N Valid	Mean MSE	MC SE (MSE)
Design 3	250	0.0804	0.00706
Design 8	250	0.0812	0.00713
Design 7	250	0.1071	0.00975
Design 6	250	0.1073	0.00983
Design 4	250	0.1465	0.01275
Design 5	250	0.2237	0.02064
Design 2	250	0.2258	0.02124
Design 1	250	0.7834	0.05582

19 Appendix A: Simulation Pseudocode

Revised structure (2026-09). Every random draw is preceded by `set.seed(digest2int(key))` with the key shown; $K = 10$ surfaces, $J = 25$ design draws per surface.

```
BUILD spatial_grid -> coords, W_rook, W_queen
```

```
FOR EACH incidence_config (mode, rho_X) IN {iid; spatial 0.20, 0.50; poisson 0.20, 0.50}:
```

```
  FOR k IN 1:K:
```

```
    GENERATE surface X_k
```

```
    # key "X", mode, rho_X, k
```

```
    # (spatial filters use W_queen; poisson P = 1)
```

```
  FOR EACH nb_type IN {rook, queen}:
```

```
    FOR EACH rho IN {0.00, 0.01, 0.20, 0.50}:
```

```
      COMPUTE inv_mat = solve(I - rho * W)
```

```
      FOR EACH gamma IN {0.50, 0.60, 0.70, 0.80}:
```

```
        FOR EACH spill_type IN {control_only, both}:
```

```
          # one "block"
```

```
          GENERATE Epsilon (N x K*J) ~ N(0, 1)
```

```
          # key "eps", mode, rho_X, nb, rho, gamma,
```

```
          # shared by all 8 designs and all tau
```

```
    FOR EACH design d IN 1:8:
```

```
      FOR k IN 1:K:
```

```

GENERATE Z_k1..Z_kJ from X_k          # key "Z", mode, rho_X, nb, rho, gamma, sp
                                     # (Checkerboard: same Z every time)
FOR EACH tau IN {0.8, 1.0, 1.5, 2.0, 3.0}:
  FOR each fit (k, j):
    Y = inv_mat %*% (tau*Z_kj + Spill(Z_kj) + beta*X_k + eps_kj)
    FIT oracle Y ~ Z + Spill + X_k      # fit_sar_lag(); flag aliased fits
    FIT non-oracle Y ~ Z + X_k          # same Y
  COMPUTE Bias, SD, MSE, Coverage, Power, Fail_Rate over the 250 fits
  COMPUTE surface-level SEs from the K per-surface metrics
  STORE one results row per estimator

```

20 Appendix B: File Locations

Table 38: RDS result files

File	Location
run_info_20260924_025509.rds	results/sim_data/
sim_results_DIM_combined_20260304_195321.rds	results/sim_data/
sim_results_DIM_iid_20260304_195321.rds	results/sim_data/
sim_results_DIM_poisson_20260304_195321.rds	results/sim_data/
sim_results_DIM_spatial_20260304_195321.rds	results/sim_data/
sim_results_MLE_tau_sweep_combined_20260924_025509.rds	results/sim_data/
sim_results_MLE_tau_sweep_iid_20260924_025509.rds	results/sim_data/
sim_results_MLE_tau_sweep_poisson_20260924_025509.rds	results/sim_data/
sim_results_MLE_tau_sweep_spatial_20260924_025509.rds	results/sim_data/
sim_results_MLEnonoracle_tau_sweep_combined_20260924_025509.rds	results/sim_data/
sim_results_MLEnonoracle_tau_sweep_iid_20260924_025509.rds	results/sim_data/
sim_results_MLEnonoracle_tau_sweep_poisson_20260924_025509.rds	results/sim_data/
sim_results_MLEnonoracle_tau_sweep_spatial_20260924_025509.rds	results/sim_data/
surface_results_MLE_tau_sweep_20260924_025509.rds	results/sim_data/
surface_results_MLEnonoracle_tau_sweep_20260924_025509.rds	results/sim_data/

All simulation data files are located in: `results/sim_data/`

Code modules: `code/01_spatial_setup.R` through `code/08_design_recommendations.R`

Reports: `code/00_mathematical_specification.Rmd`, `code/07_results_summary.qmd`,
`code/09_MLE_design_recommendation_report.qmd`